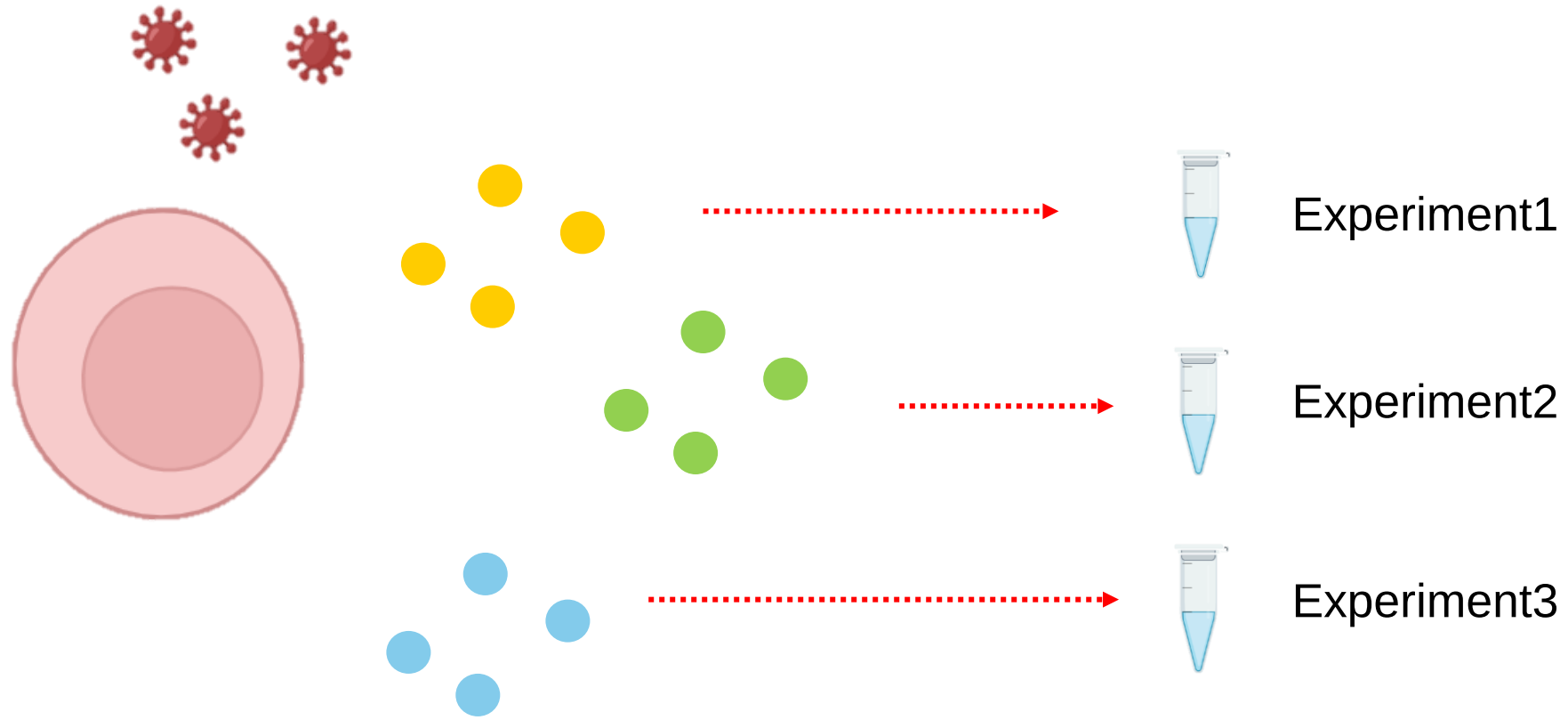
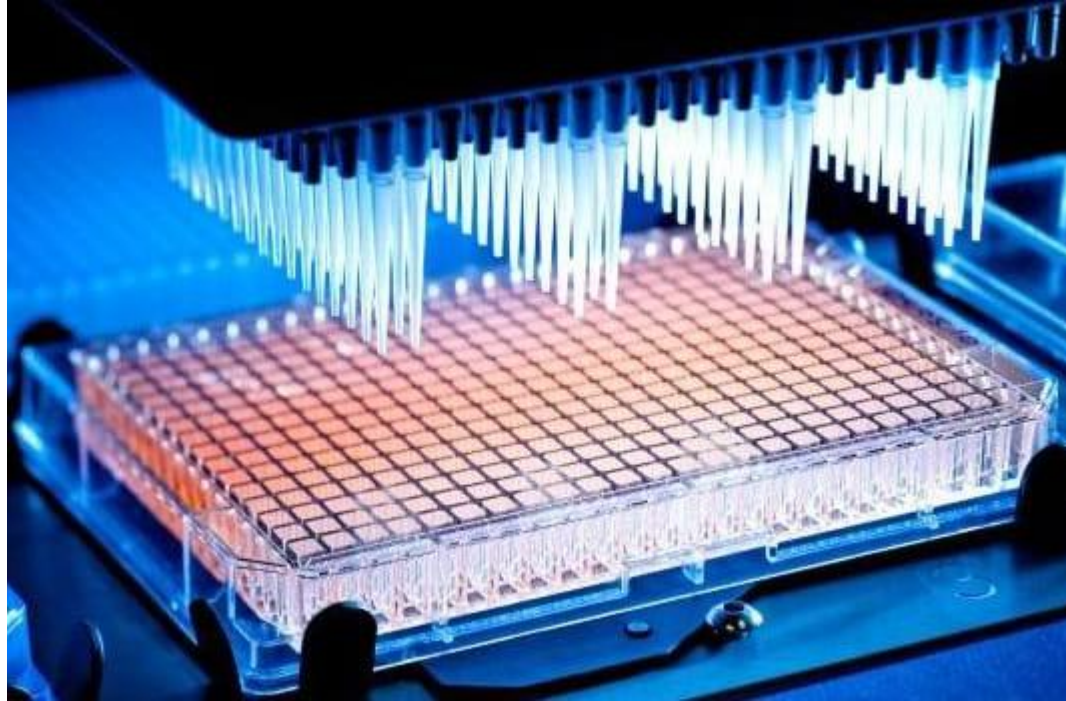
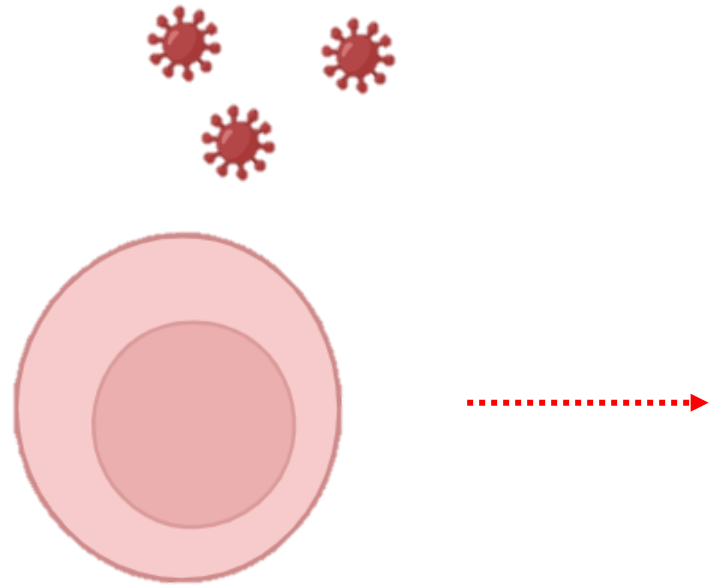


Bulk RNA-sequencing

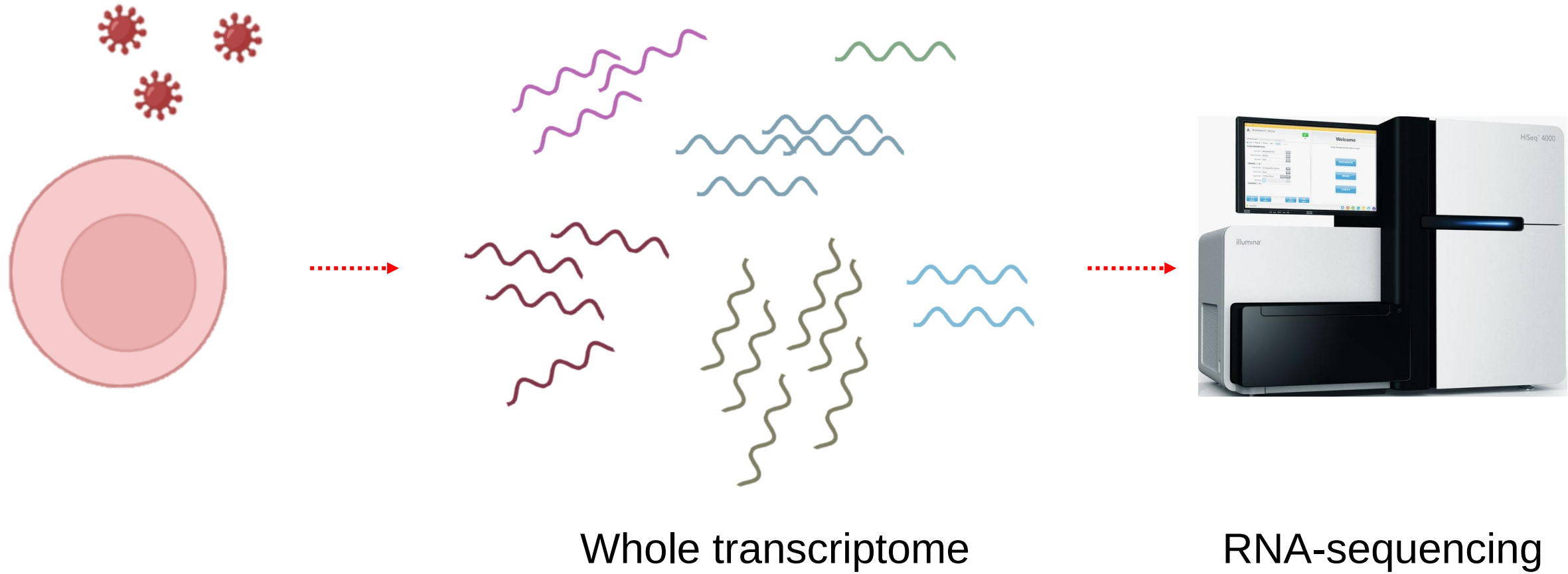
- Data acquisition



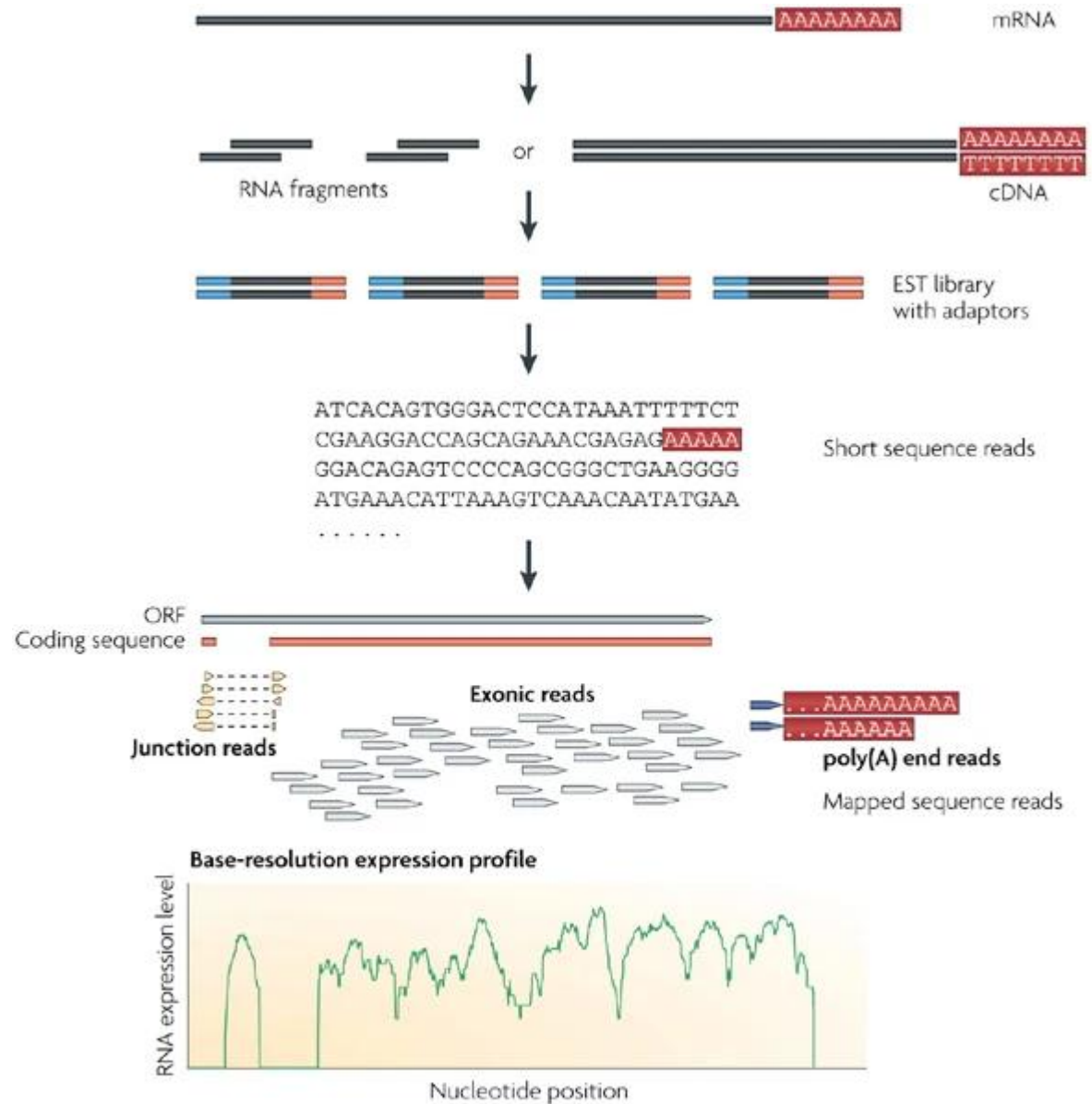
- High throughput data



- High throughput data



• RNA-sequencing

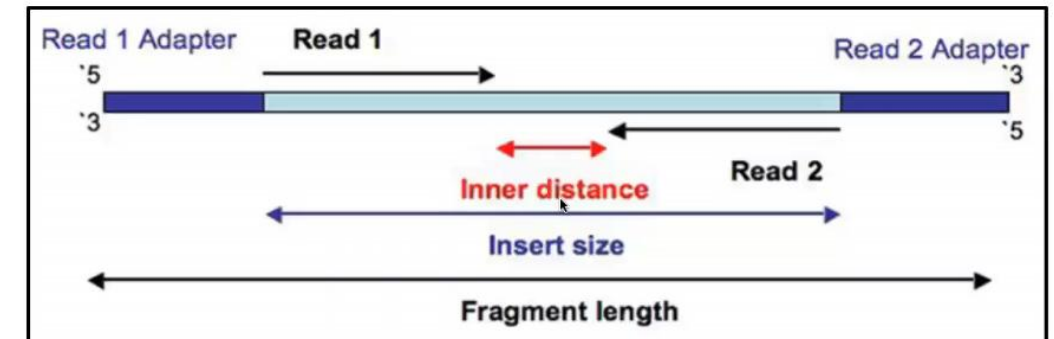


- Raw mRNA
- RNA fragment chopping
- Reverse transcriptase
- Adaptors → Obtain the sequence

Read (single-end) (100 ~ 200 bp)

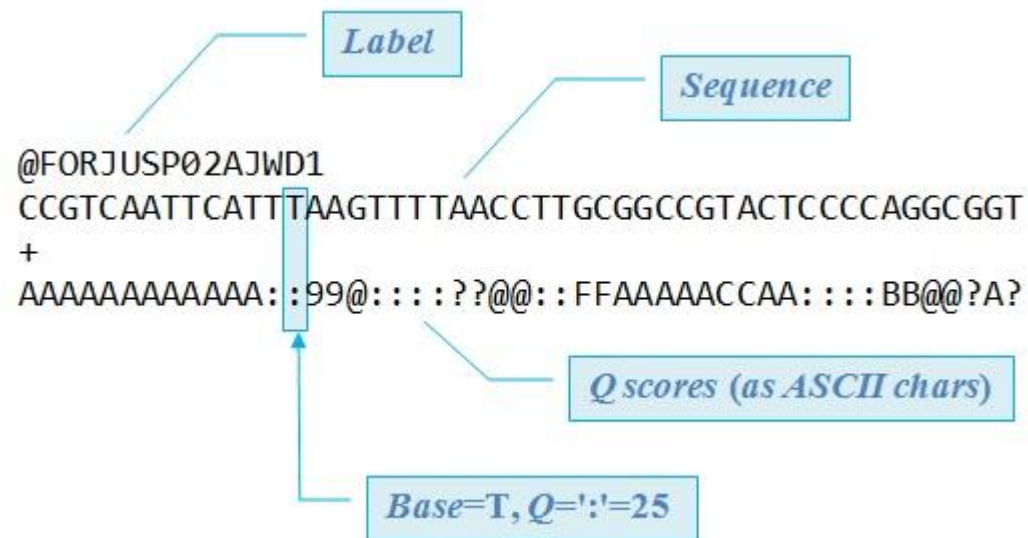
Or

Fragment (paired-end): high confidence



• RNA-sequencing

Data format: fastq file



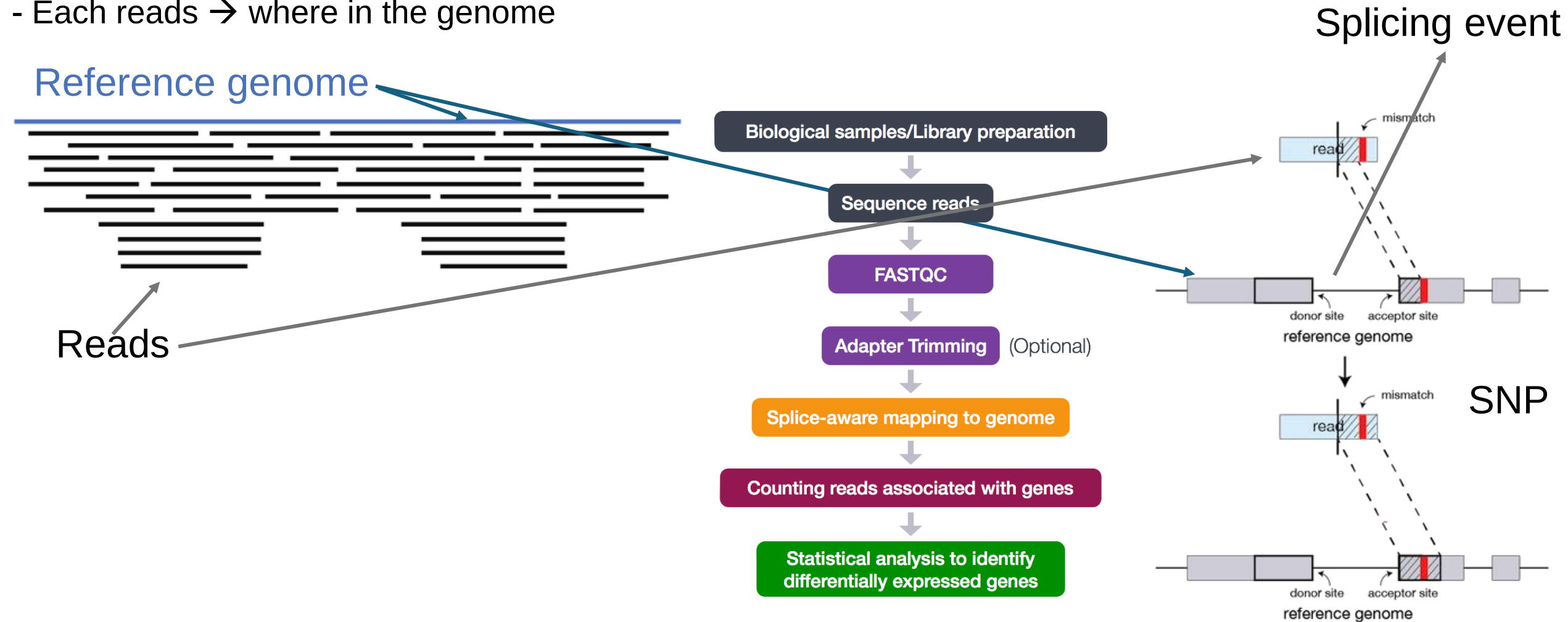
```
@A01001:23:HKJ3JDRXX:1:1101:1307:1000 2:N:0:GATTAGAT
TCTGACCCTTTTCCACAGGGGACCTACCCCTATTGCGGTCCTCCAGCTCATCTTTCACCTCACCCCCCTCCTCCTTGGCTTTAAT
+
FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF,FFF:FFFF:
@A01001:23:HKJ3JDRXX:1:1101:1325:1000 2:N:0:CTGACTGA
GCAGTGGTATCAACGCAGAGTACATGGGAATAACGCCGCCGCATCGCCGGTCGGCATCGTTTATGGTCGGAACACGACGGTATCTGAT
+
FFFFFFFFF:FFF:FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF,FFFFFFFFF:FFFFFFFFFFFFFFFFF
@A01001:23:HKJ3JDRXX:1:1101:1344:1000 2:N:0:ACCGTATG
ATAGGCTAGTGTGGGATTGCTCCACCCAGAGGCCCTTCCCCAGAGCAGGGAGGACATGGAGTGTGTGAAGGTTTTCTCTCCTTAAC
+
FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF,FFFFFFFFF,FFFFFFFFFFFFFFFFF
@A01001:23:HKJ3JDRXX:1:1101:1542:1000 2:N:0:GATTAGAT
AAGCAGTGGTATCAACGCAGAGTACATGAGAAGTGCCCCACCTGCTCCTCAGTTCAGCCTGACCCCTCCCATCCTTTGGCCTCTGA
+
FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF,FFFFFF:FFFF:FFFF:FFF:FFFFFFFFFFFFF,FFF::FFFFFFFFFFFFFFFFF
@A01001:23:HKJ3JDRXX:1:1101:1561:1000 2:N:0:TGACGCC
AAACTTATGAAGATCAGGAAATTTACCTATATTCAAAGAAAAGAAATTAATGAAACCAGCTGTGAAATTACTCAGATGTTGAAA
```

• RNA-sequencing (Alignment)

Fastq file → somehow **gene by count** matrix (it could also be a transcript or isoform)

1) Alignment (ex: STAR → Genome, Kallisto → Transcriptome)

- Each reads → where in the genome



• RNA-sequencing (Alignment)

- Genome: different genome assembly version
Human: GRCh37.## (hg19), GRCh38.## (hg38)
Mouse: GRCm38, GRCm39
Different versions cannot be used together
→ Different nucleotide locus
→ Same genome build but different version
(or release): compatible with each other
(mostly gap-filling)

- Human genome size: 3.1 Gbase pairs, Mouse genome size: 2.7Gbp → mapping is not trivial
Genome building: Making a dictionary for boosting the mapping time



- RNA-sequencing (Alignment)

Read1, 2, 3 ... → Gene1



Read13, 22, 23 ... → Gene2



Read37, 211, 309 ... → Gene3

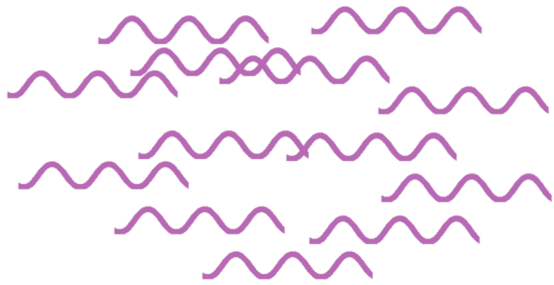


Output: SAM or BAM file (BAM: binary file)

Started job on	Jul 17 20:54:45
Started mapping on	Jul 17 20:55:04
Finished on	Jul 17 21:06:18
Mapping speed, Million of reads per hour	160.92
Number of input reads	30128333
Average input read length	202
UNIQUE READS:	
Uniquely mapped reads number	27974985
Uniquely mapped reads %	92.85%
Average mapped length	201.55
Number of splices: Total	22952060
Number of splices: Annotated (sjdb)	22816849
Number of splices: GT/AG	22776111
Number of splices: GC/AG	145758
Number of splices: AT/AC	18027
Number of splices: Non-canonical	12164
Mismatch rate per base, %	0.17%
Deletion rate per base	0.01%
Deletion average length	1.92
Insertion rate per base	0.01%
Insertion average length	1.50
MULTI-MAPPING READS:	
Number of reads mapped to multiple loci	1262485
% of reads mapped to multiple loci	4.19%
Number of reads mapped to too many loci	10600
% of reads mapped to too many loci	0.04%
UNMAPPED READS:	
Number of reads unmapped: too many mismatches	0
% of reads unmapped: too many mismatches	0.00%
Number of reads unmapped: too short	874728
% of reads unmapped: too short	2.90%
Number of reads unmapped: other	5535
% of reads unmapped: other	0.02%
CHIMERIC READS:	
Number of chimeric reads	0
% of chimeric reads	0.00%

- RNA-sequencing (Alignment)

Unique mapped read → what we use for data analysis
What are other reads?



Human read



Microbiome reads,
contamination ...



Long read: High specificity → won't map to other regions
Ex): PacBio; high error rate



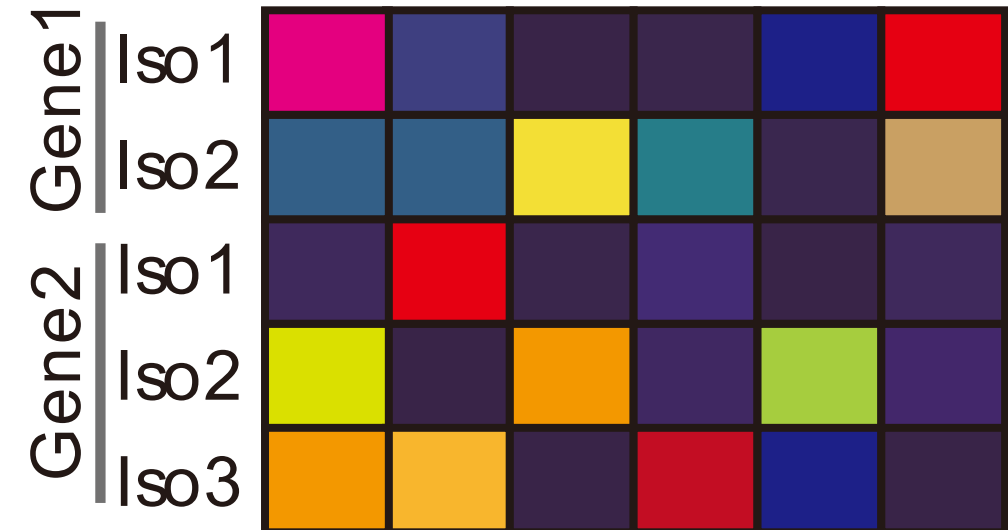
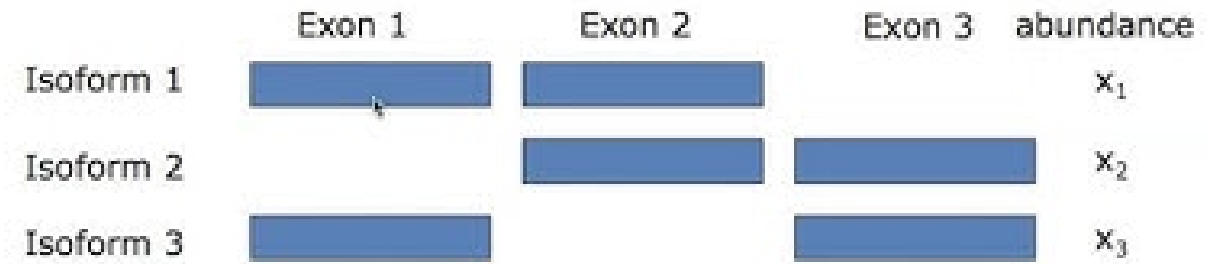
Short read: Multiple mapping to many regions

• RNA-sequencing (Quantification)

- Mapped reads → count matrix (per gene)
Counting reads mapped to a given gene
Ex) FeatureCounts (gene), Kallisto (transcript)

	Cell1	Cell2	...	CellN
Gene1	3	2	.	13
Gene2	2	3	.	1
Gene3	1	14	.	18
...	.	.	.	.
...	.	.	.	.
...	.	.	.	.
GeneM	25	0	.	0

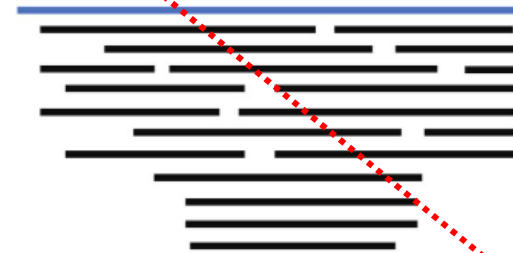
- Mapped reads → count matrix (per isoform)
Unique exon count, EM (expectation maximization) algorithm
Ex) RSEM or longread sequencing



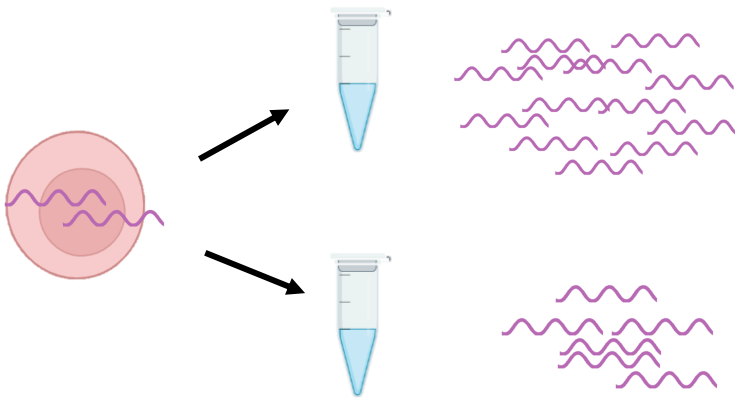
• RNA-sequencing (Normalization)

- Why?

1) Gene length normalization: Different probability of “read” capture rate during sequencing
→ longer gene has a higher chance of being mapped



2) Total read count (= read-depth) normalization: Adjust read-depth between different samples



Should be the same

3) RPKM (Reads Per Kilobase per Millions mapped reads), FPKM (Fragment), TPM (Transcript)

$$\text{RPKM} = \frac{\text{Total fragments}}{\text{Mapped reads (millions)} * \text{exon length (KB)}} = \frac{\text{Number of reads of the region}}{\frac{\text{Total reads}}{1,000,000} * \frac{\text{Region length}}{1,000}}$$

TPM: Total reads normalization → total read after gene length normalization

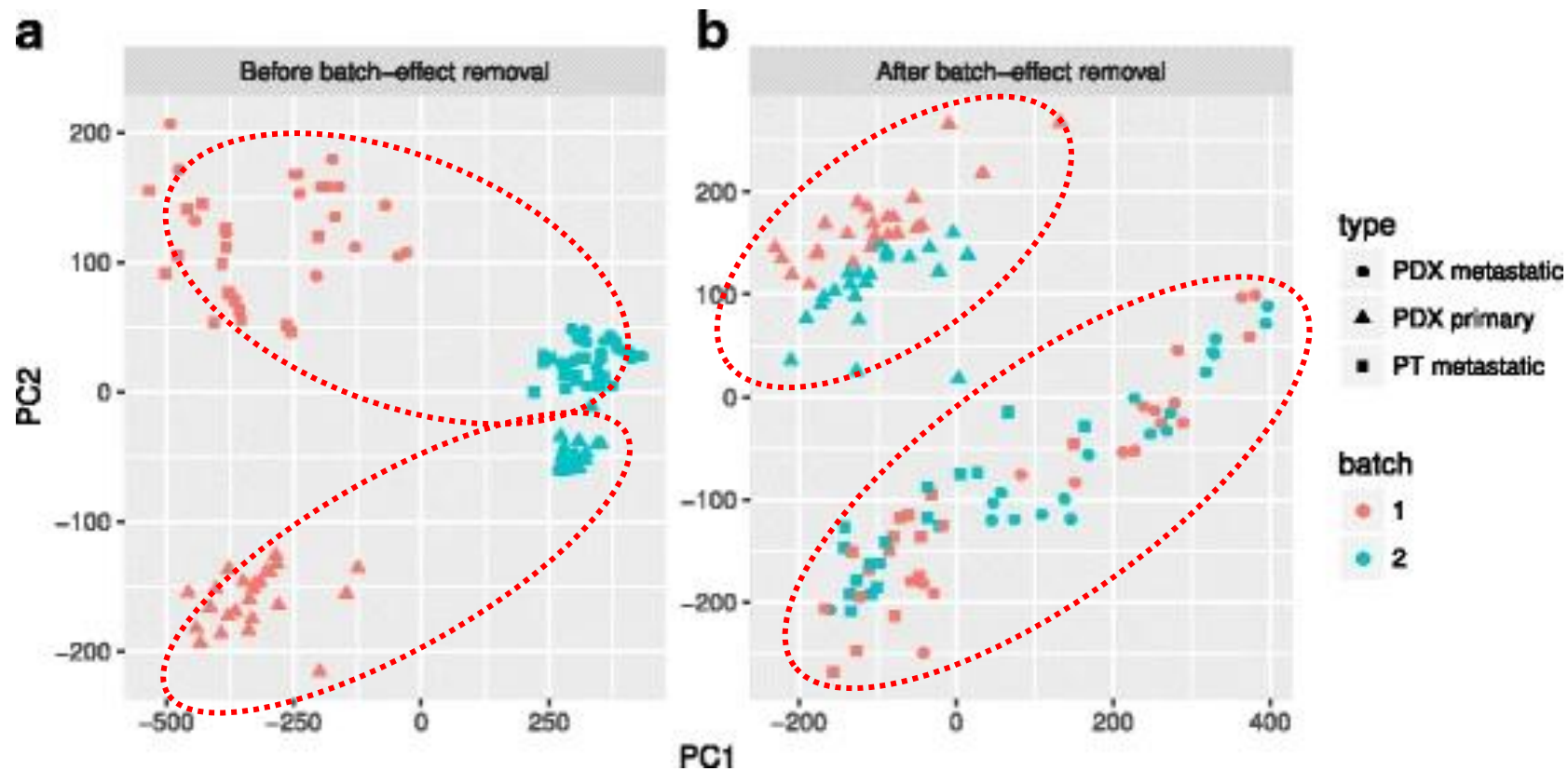
- RNA-sequencing (Batch correction)

- Why?

Same sample → but, a technical confounding effect

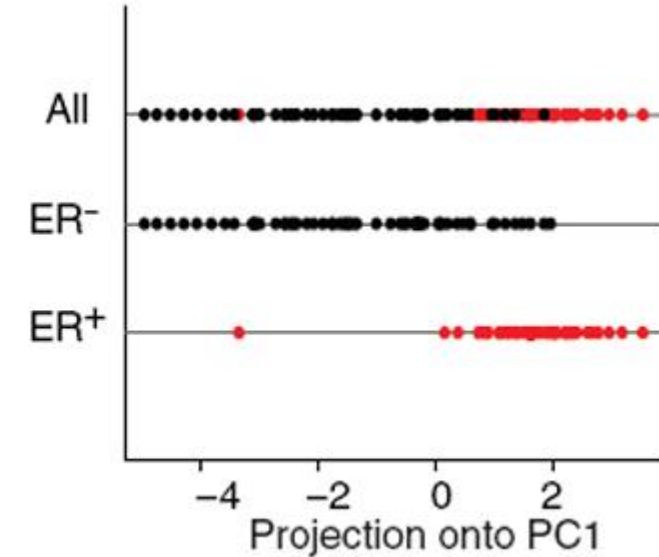
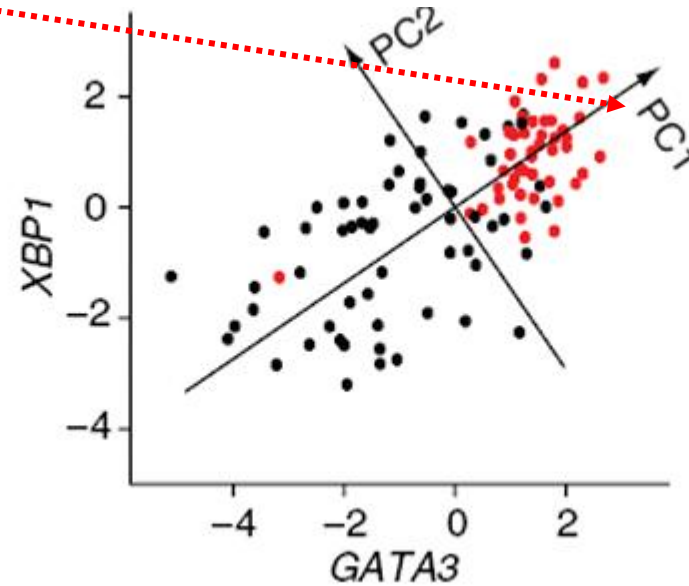
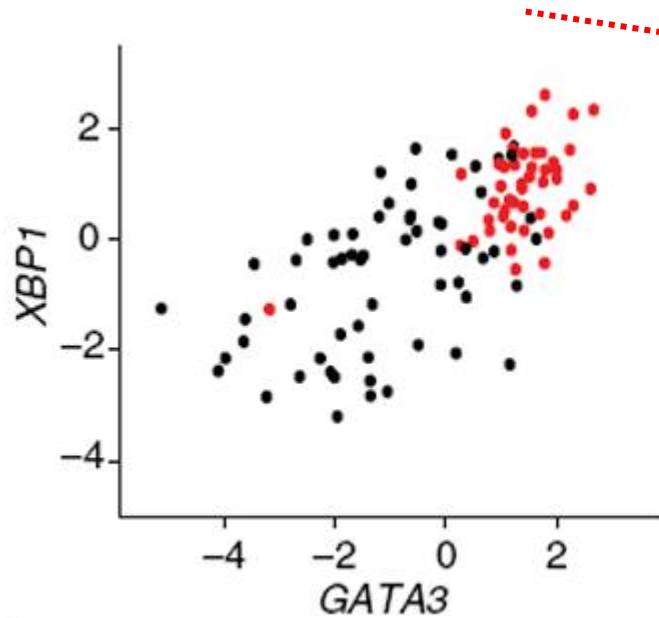
(My experiment and your experiment should be the same!)

*Limma: linear regression-based, Combat: negative binomial distribution, DESeq2: scaling factors

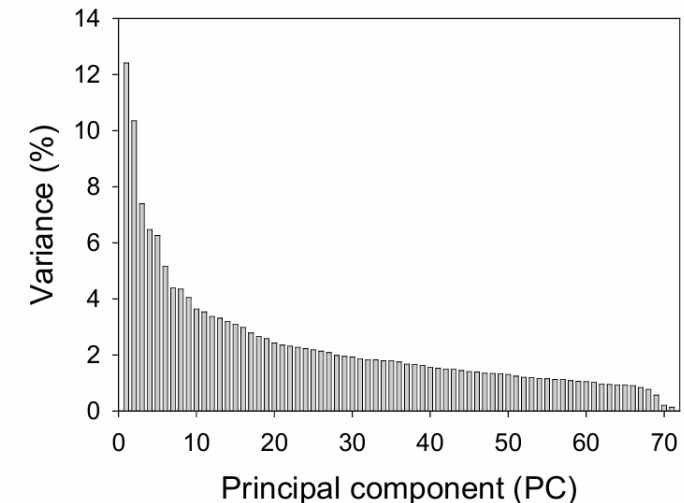


• PCA (principal component analysis)

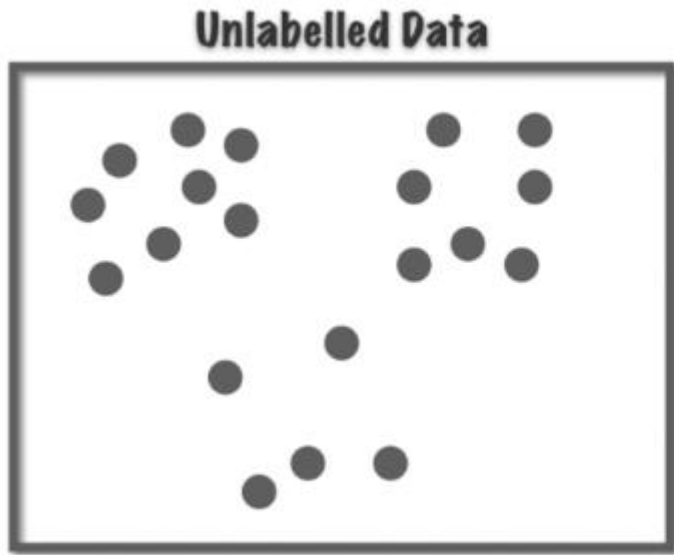
- Find a PC axis to maximize the variance of the data → Distribute samples to maximize the variance



- Number of PC == Number of features or $\leq$ sample size
 - Usually, we have more features (~20k)
 - Each PCs: explain the variance of the data
 - Using only a few PCs: Dimension reduction
- Advantage: among confounding effects, noise, we can solely capture the biological difference (or meaningful information)
- + data representation & visualization (20K genes → 2 PCs)



- Clustering



Distance measurement for clustering

- **Euclidean distance** (a straight line distance in n -dimensional space)

$$d = \sqrt{\sum_{i=1}^n (x_i - y_i)^2}$$

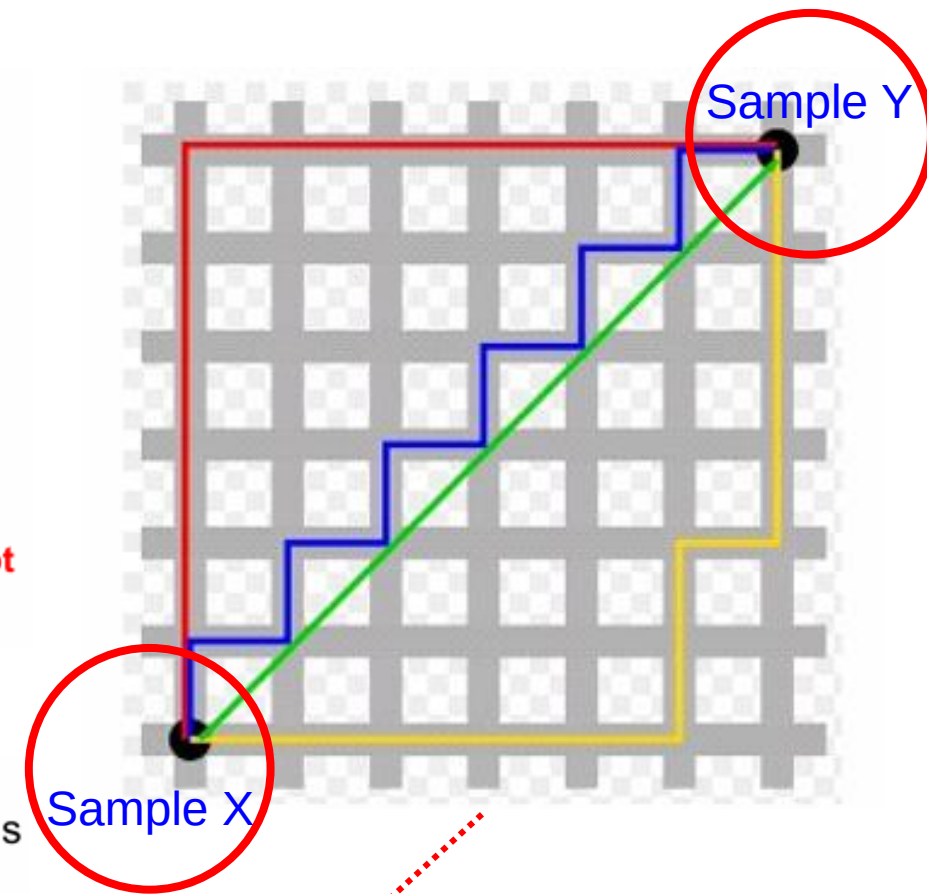
$$d = \sqrt{\frac{1}{n} \sum_{i=1}^n (x_i - y_i)^2}$$
 ,if you don't want to increase the distance with the addition of more dimensions.

- Euclidean distances **may underestimate** join differences such as differences in two correlated expression .
- Therefore, **use Euclidean distance if you believe that your dimensions (variables) are not independent.**

- **Manhattan distance** (= city block distance, L1 distance, rectilinear distance, taxicab metric): distances measured parallel to dimensional axes. After NY city's grid like street pattern.

$$d = \sum_{i=1}^n |x_i - y_i|$$

$$d = \frac{1}{n} \sum_{i=1}^n |x_i - y_i|$$
 ,if you don't want to increase the distance with the addition of more dimensions.

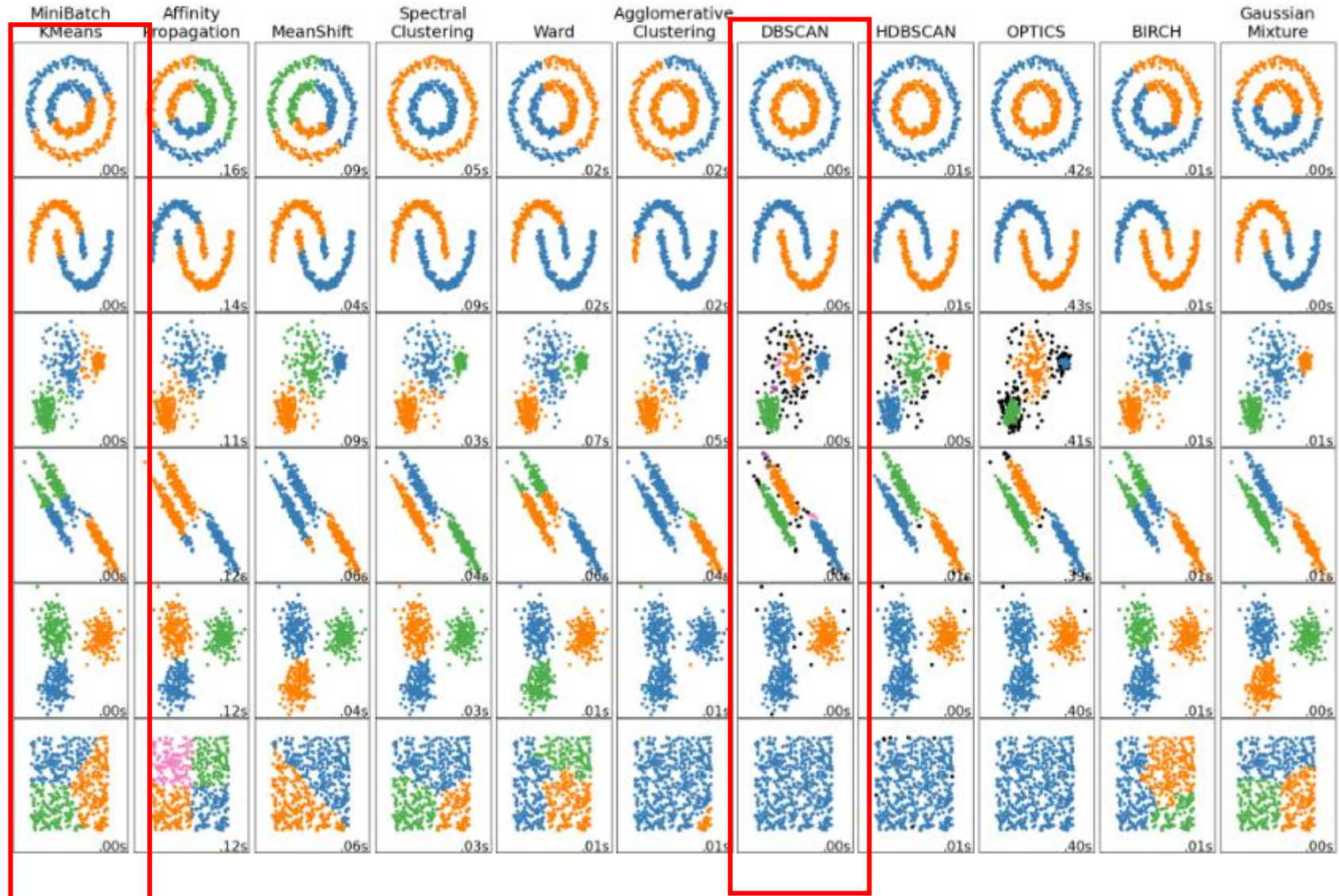


Gene expression space

i: each gene

n: the number of genes

- Clustering

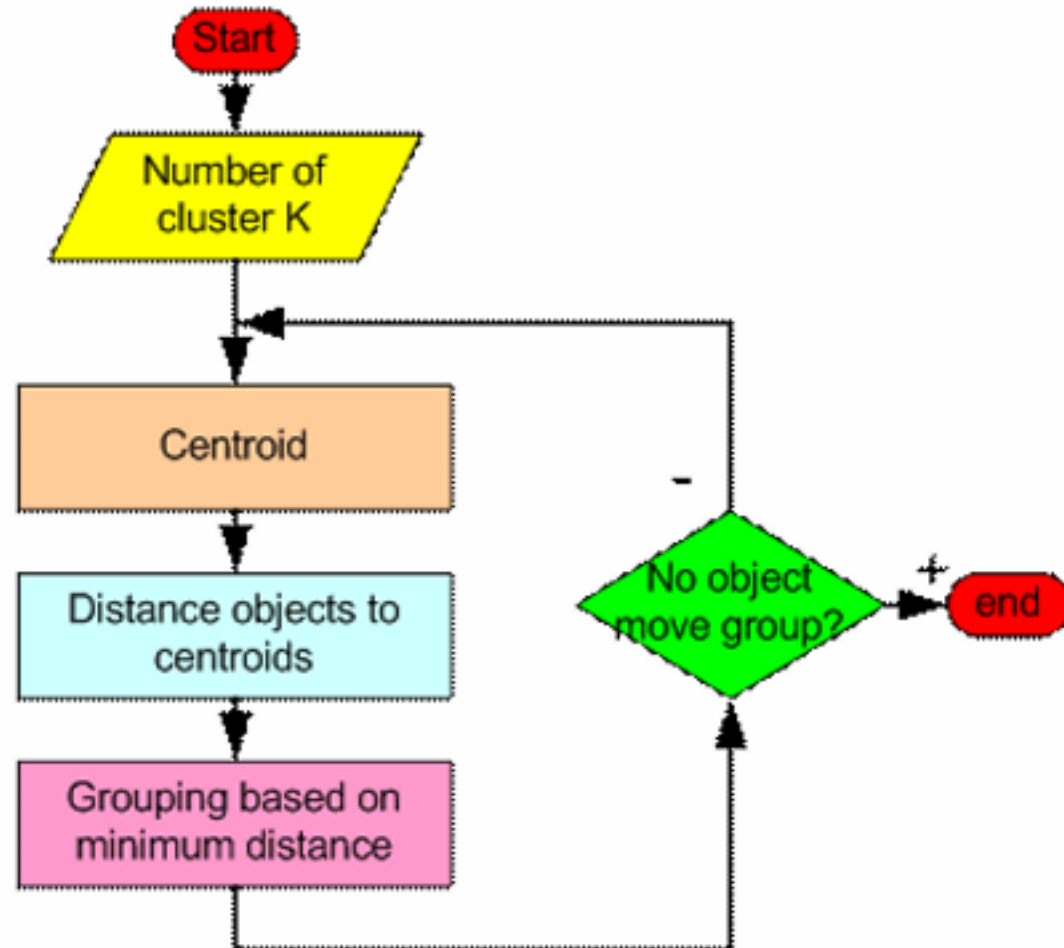


- K-mean clustering

K-mean clustering (linear approach)

➤ **Algorithm**

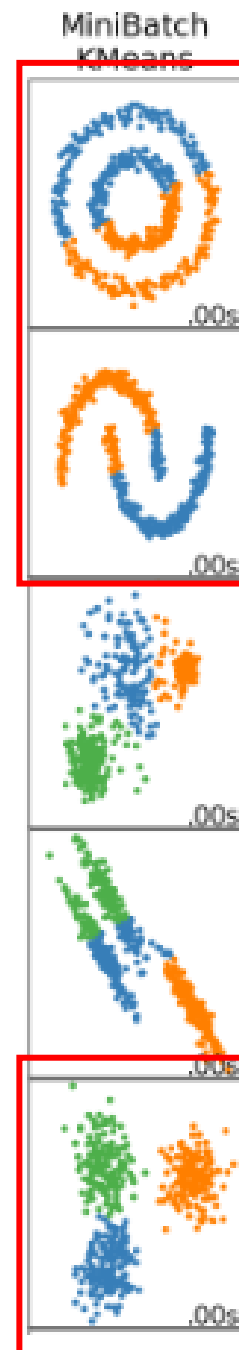
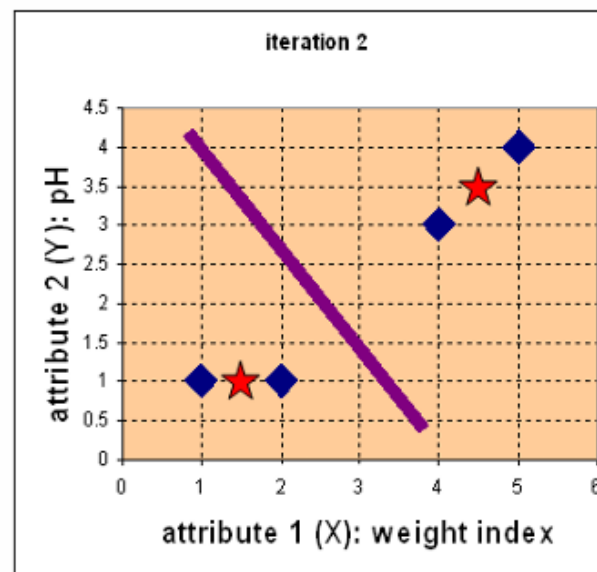
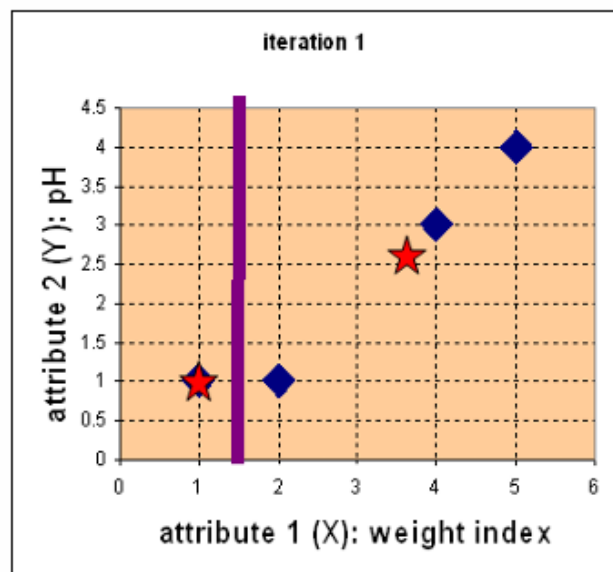
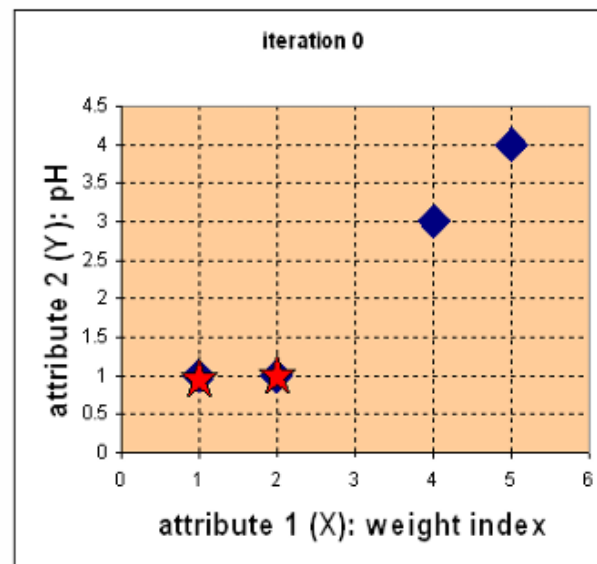
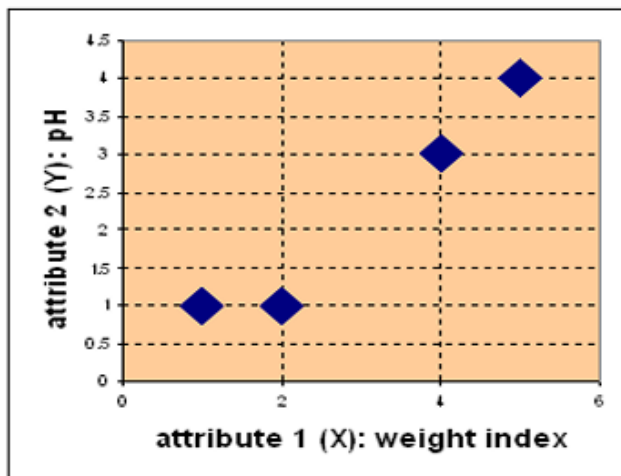
1. **Initial k :** Decide the number of cluster (k)
2. **Initial value of centroids:** Choose k centroids randomly.
3. **Objects-centroids distance:** Calculate the distance between cluster centroid to each object (with any distance metric).
4. **Objects clustering:** Assign each object based on minimum distance.
5. **Determine new centroids:** New centroid moves to mean value of all member objects.
6. **Iterate steps 3-5:** Until no object change centroid membership.



- K-mean clustering

Demonstration

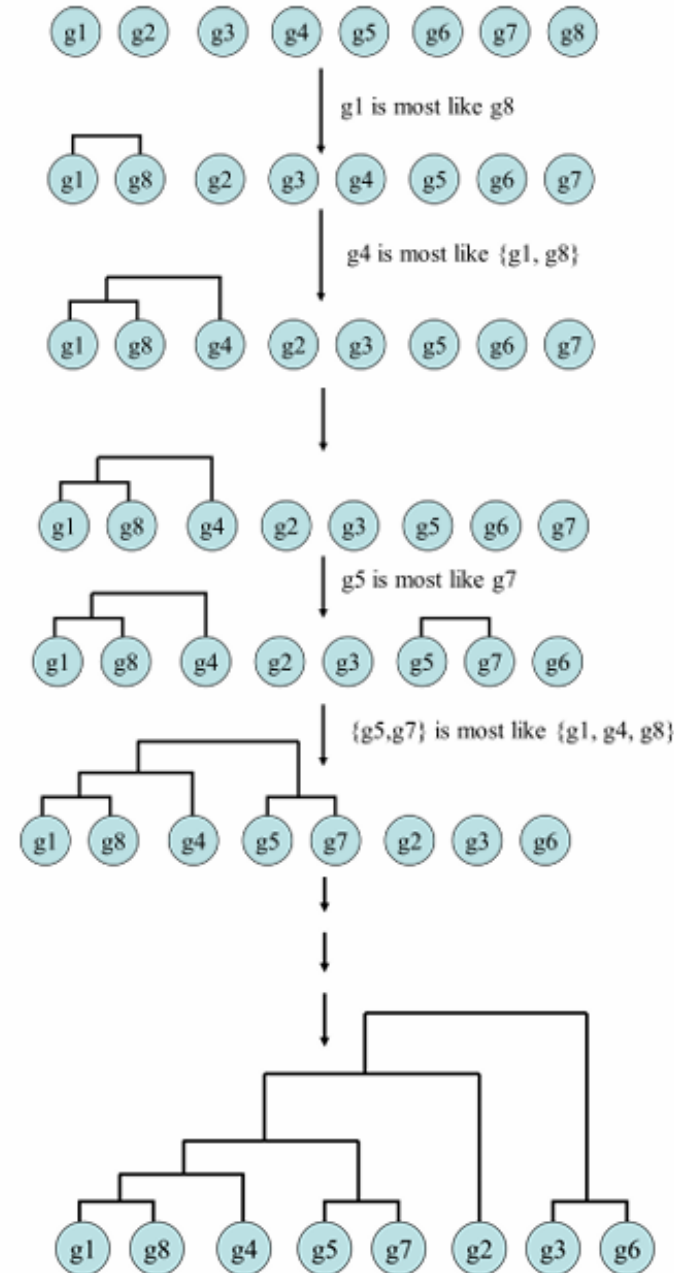
Diamond: object
Star: centroid ($k=2$)



• Hierarchical clustering

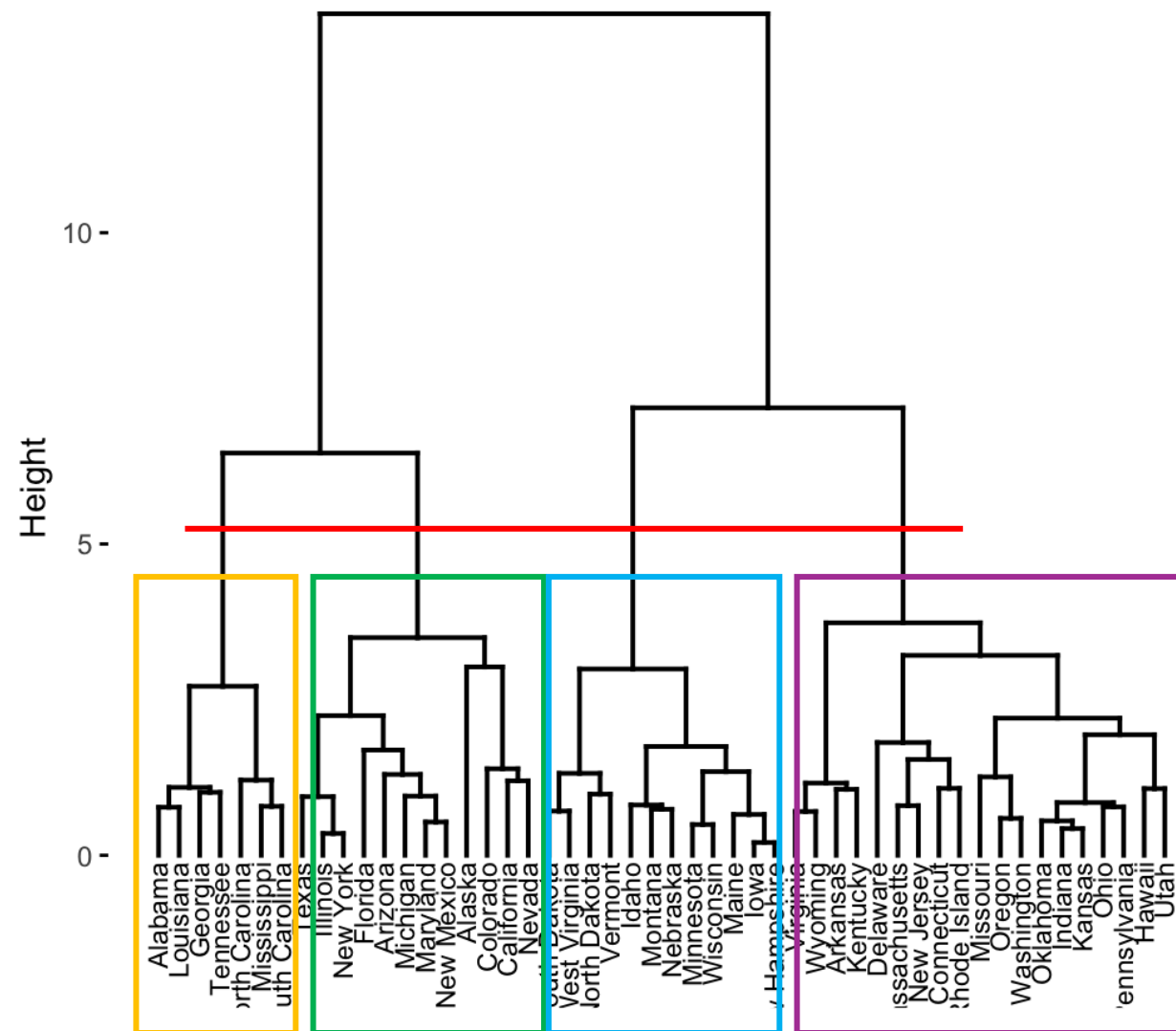
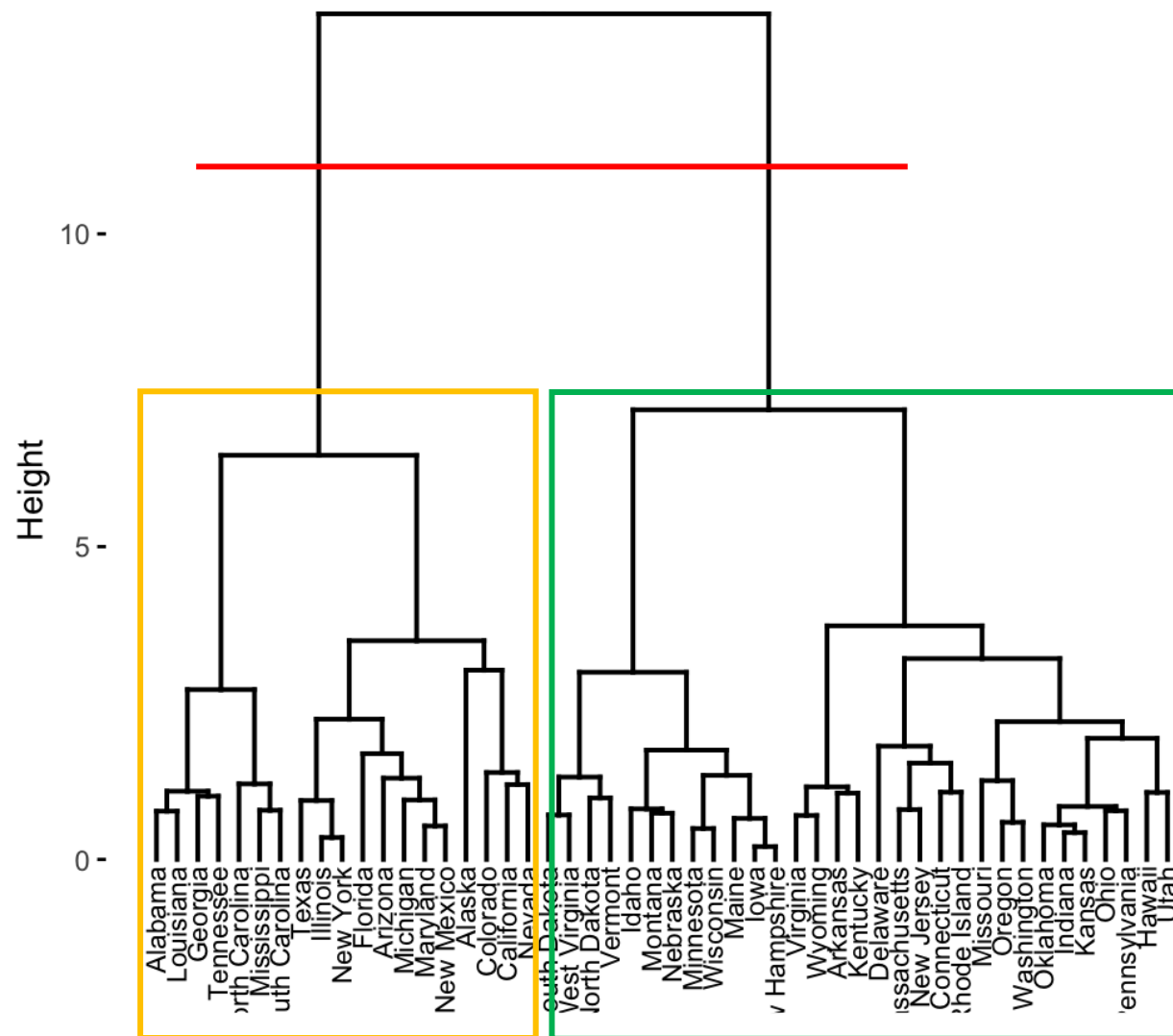
➤ Algorithm

1. Initialize each single gene as a cluster
2. The pairwise distance matrix is calculated for all of the genes to be clustered.
3. The distance matrix is searched for the two most similar genes or clusters.
4. The two selected clusters are merged to produce a new cluster that now contains at least two objects.
5. The distances are calculated between this new cluster and all other clusters. There is no need to calculate all distances as only those involving the new cluster have changed.
6. Steps 3-5 are repeated until all objects are in one cluster.



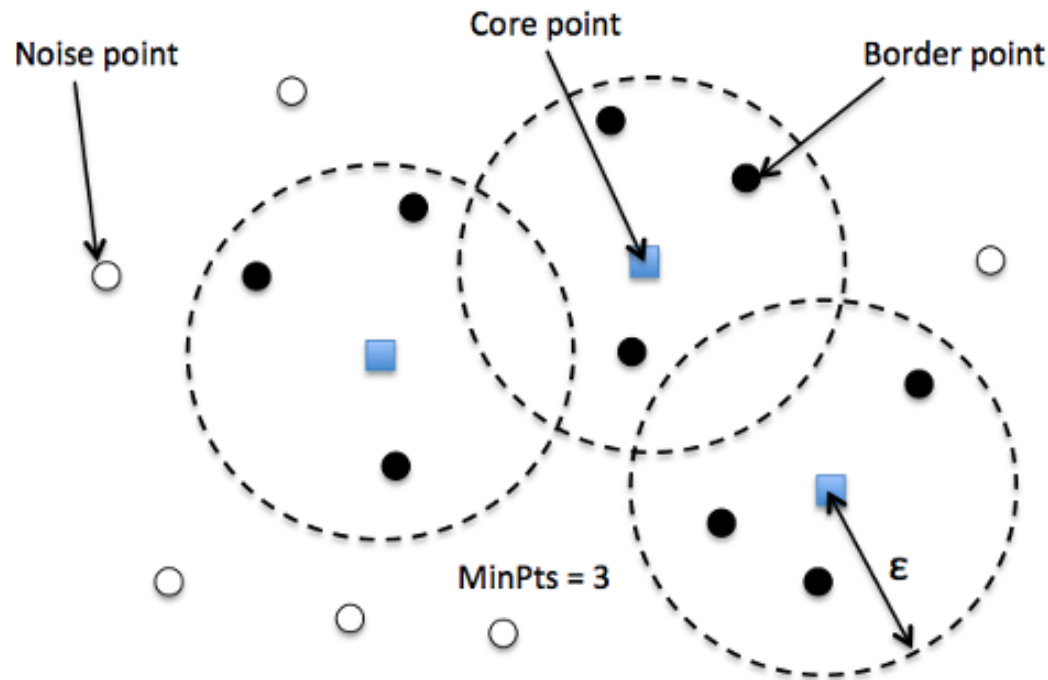
- Hierarchical clustering

Tree cutting



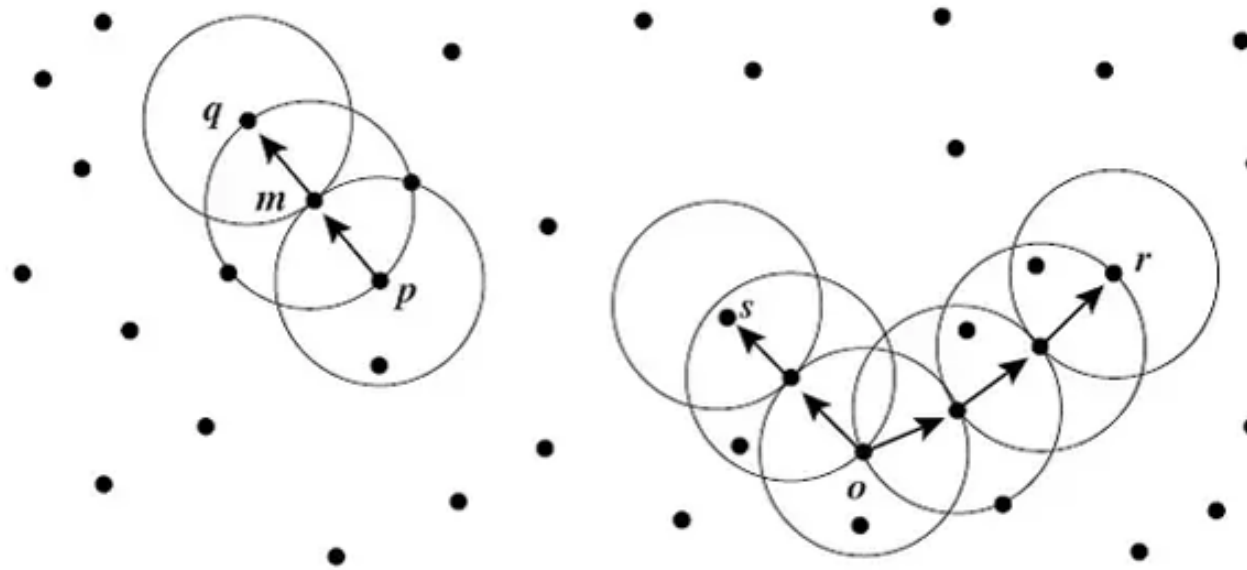
• DBSCAN (Density-Based Spatial Clustering of Applications with Noise)

- Clustering algorithm that divides a dataset into subgroups of high density regions.
- Two parameters required for DBSCAN:
 - **Epsilon (ϵ)**: a distance parameter that defines the radius to search for nearby neighbors
 - **MinPts**: minimum number of other points required to form a cluster
- **Core point** – a point that **has at least the minPts of other points within its ϵ radius**.
- **Border point** – a point within the ϵ radius of a *core point* BUT has less than the **minPts** within its own ϵ radius
- **Noise point** – a point that is neither a *core point* or a *border point*



• DBSCAN (Density-Based Spatial Clustering of Applications with Noise)

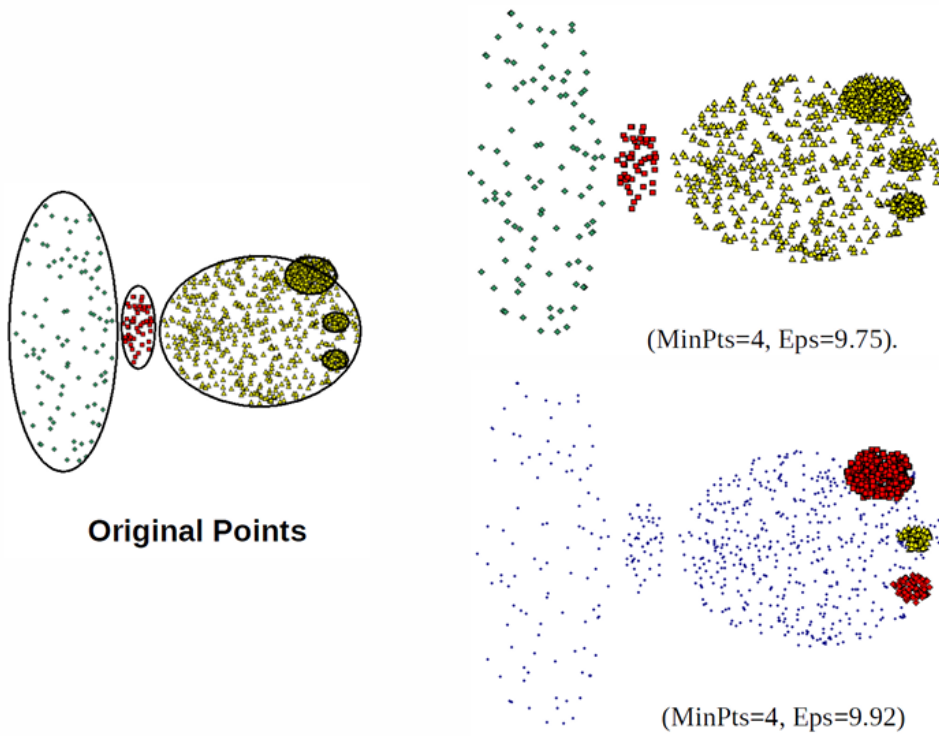
- Each core point forms a cluster together with the points that are reachable within its ϵ radius.
- **Two points are considered “directly density-reachable” if one of the points is a core point and the other point is within its ϵ radius.**
- **Larger clusters are formed when directly density-reachable points are chained together.**
- In the example image below, there are **two clusters**:



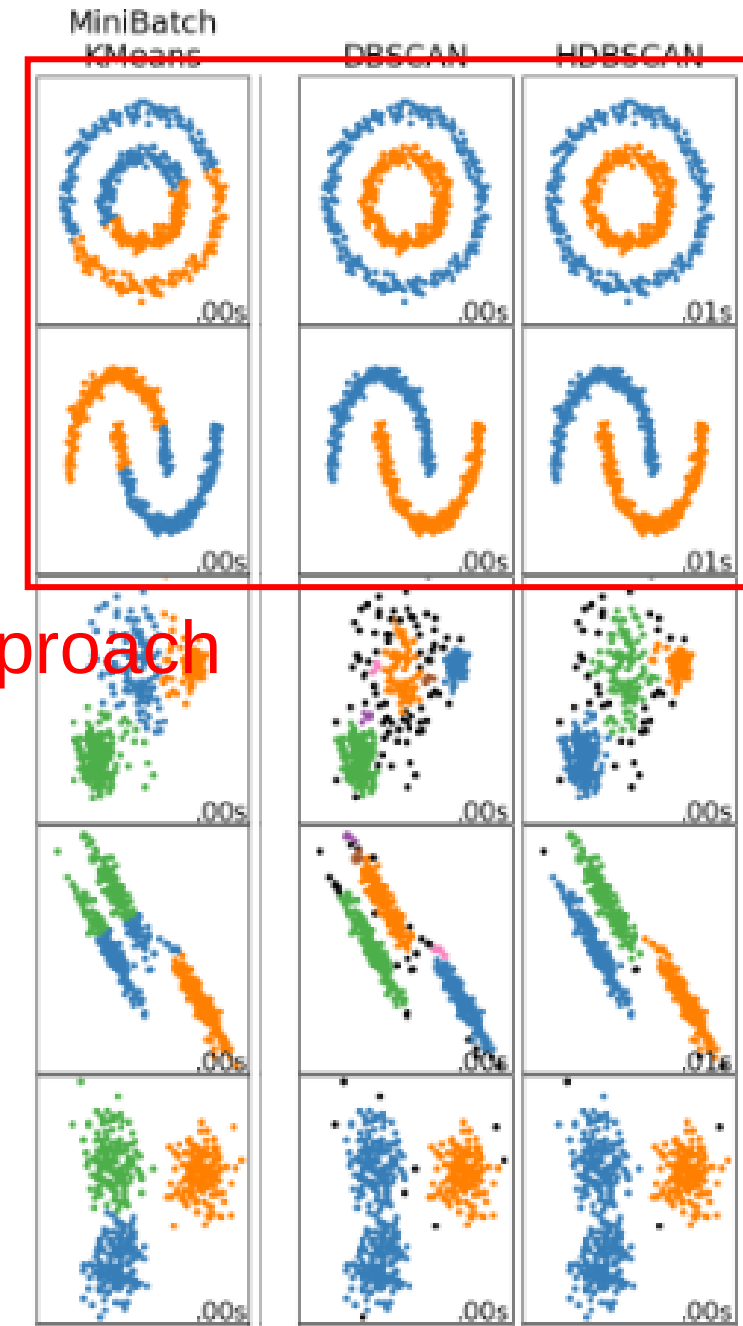
1. If minPts = 3, p is directly density-reachable from m , which is directly density-reachable from q . The sets of points within the ϵ radius of $p \rightarrow m \rightarrow q$ form one cluster
2. r and s are indirectly density-reachable through a path of 4 core points. The set of points within the ϵ radius of this chain forms another cluster.

• DBSCAN (Density-Based Spatial Clustering of Applications with Noise)

- The DBSCAN algorithm **repeats the following process until all points have been assigned to a cluster or are labeled as visited:**
 - Arbitrarily select a point P.**
 - Retrieve all points** directly density-reachable from P **with respect to ϵ .**
 - If P is a core point**, a cluster is formed. Find recursively all its density connected points and **assign them to the same cluster as P.**
 - If P is not a core point**, DBSCAN **iterates through the remaining unvisited points** in the dataset.
- DBSCAN does not require us to specify the number of clusters.
- It can handle clusters of arbitrarily shapes and sizes.
- It is robust to noise.

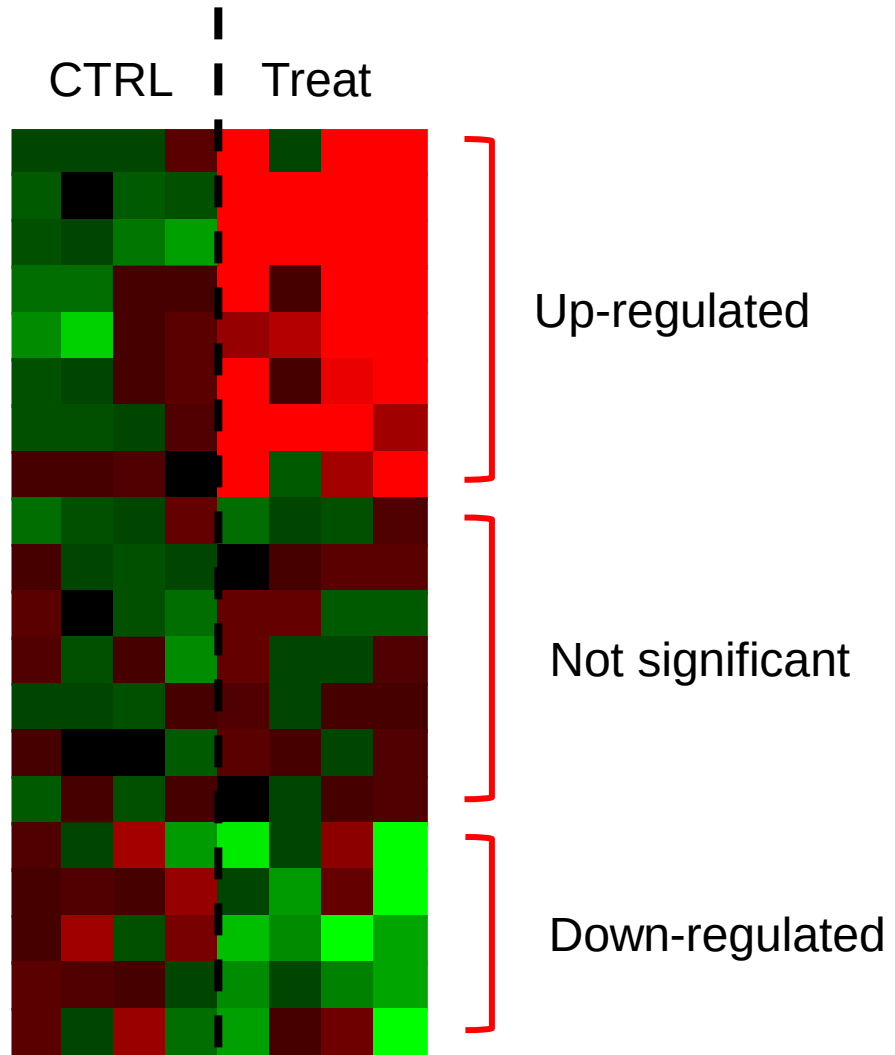


Nonlinear approach



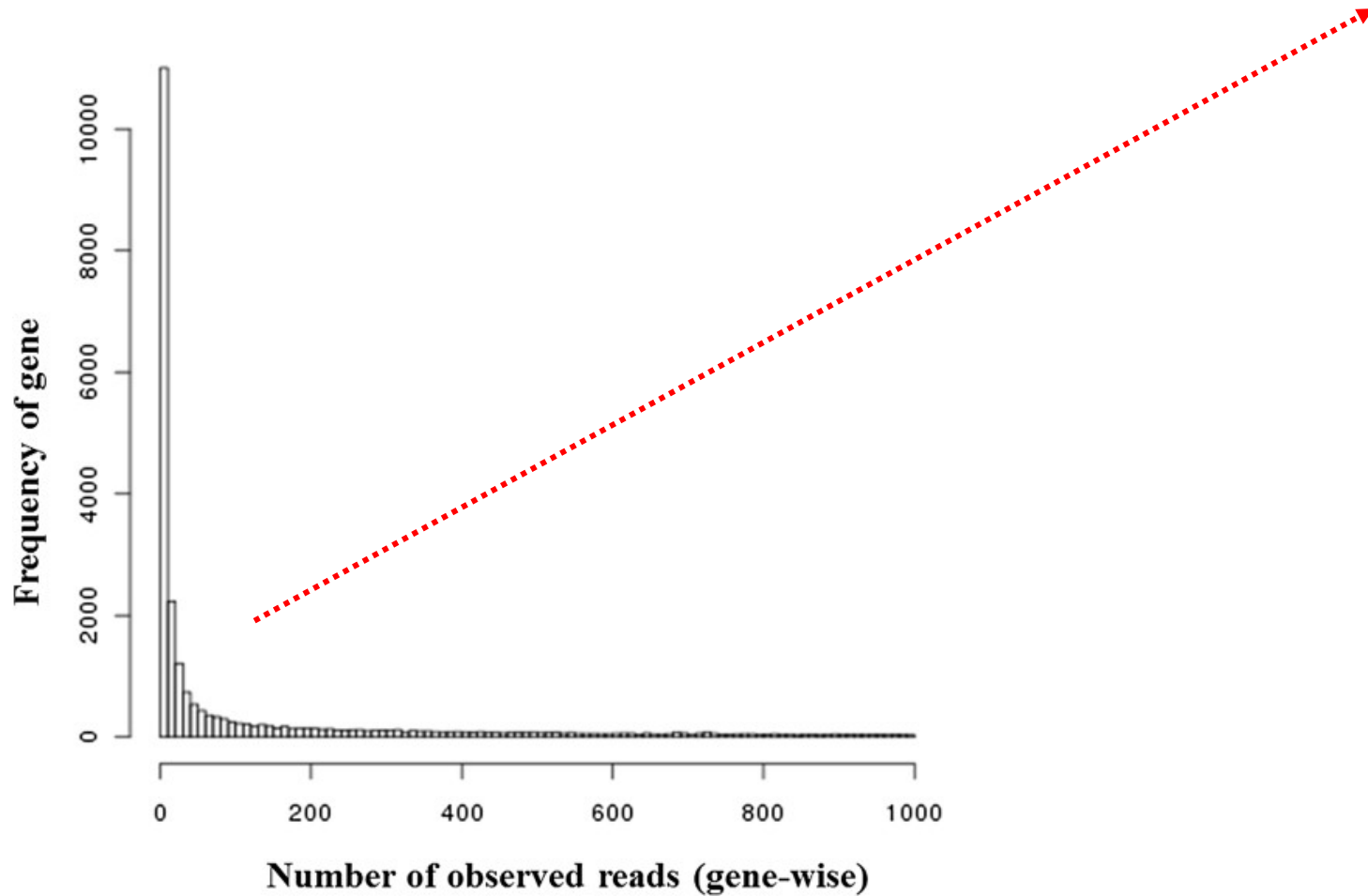
- DEG (Differentially expressed gene) analysis

-Which gene is expressing differently between two groups



- DEG (Differentially expressed gene) analysis

-DESeq2: Estimate variance-mean dependence in count data from high-throughput sequencing assays and test for differential expression based on a model using the negative binomial distribution



• DEG (Differentially expressed gene) analysis

	baseMean	log2FoldChange	lfcSE	stat	pvalue	padj	
VCAM1	4566.645595	1.938889067	0.119610186	16.21006642	4.28E-59	5.89E-55	
TNFAIP3	1099.976807	1.84009256	0.126618591	14.5325623	7.53E-48	5.18E-44	
TYMP	389.1617523	1.629446073	0.125598	12.97350338	1.73E-38	7.93E-35	
OLR1	932.8101893	1.326843941	0.105716078	12.55101369	3.92E-36	1.35E-32	
PLA2G4C	311.2901003	1.807025742	0.145769361	12.39647162	2.73E-35	7.51E-32	
BIRC3	482.0263327	4.960908659	0.402688626	12.31946555	7.12E-35	1.63E-31	
NFKBIE	235.0653324	1.858746559	0.153075883	12.14264794	6.28E-34	1.23E-30	
IL34	138.1967553	3.069325655	0.256225942	11.97898086	4.58E-33	7.87E-30	
NFKBIA	1500.794017	1.422924741	0.129867476	10.95674439	6.17E-28	9.42E-25	
RELB	460.3645001	1.895788818	0.17429306	10.8770184	1.48E-27	2.04E-24	
TRIM47	488.2175048	1.579057579	0.147620709	10.69672126	1.05E-26	1.32E-23	

Log2(treat / ctrl)
→ 0: no change
→ +: increase, -: decrease
→ Value 1: 2 fold

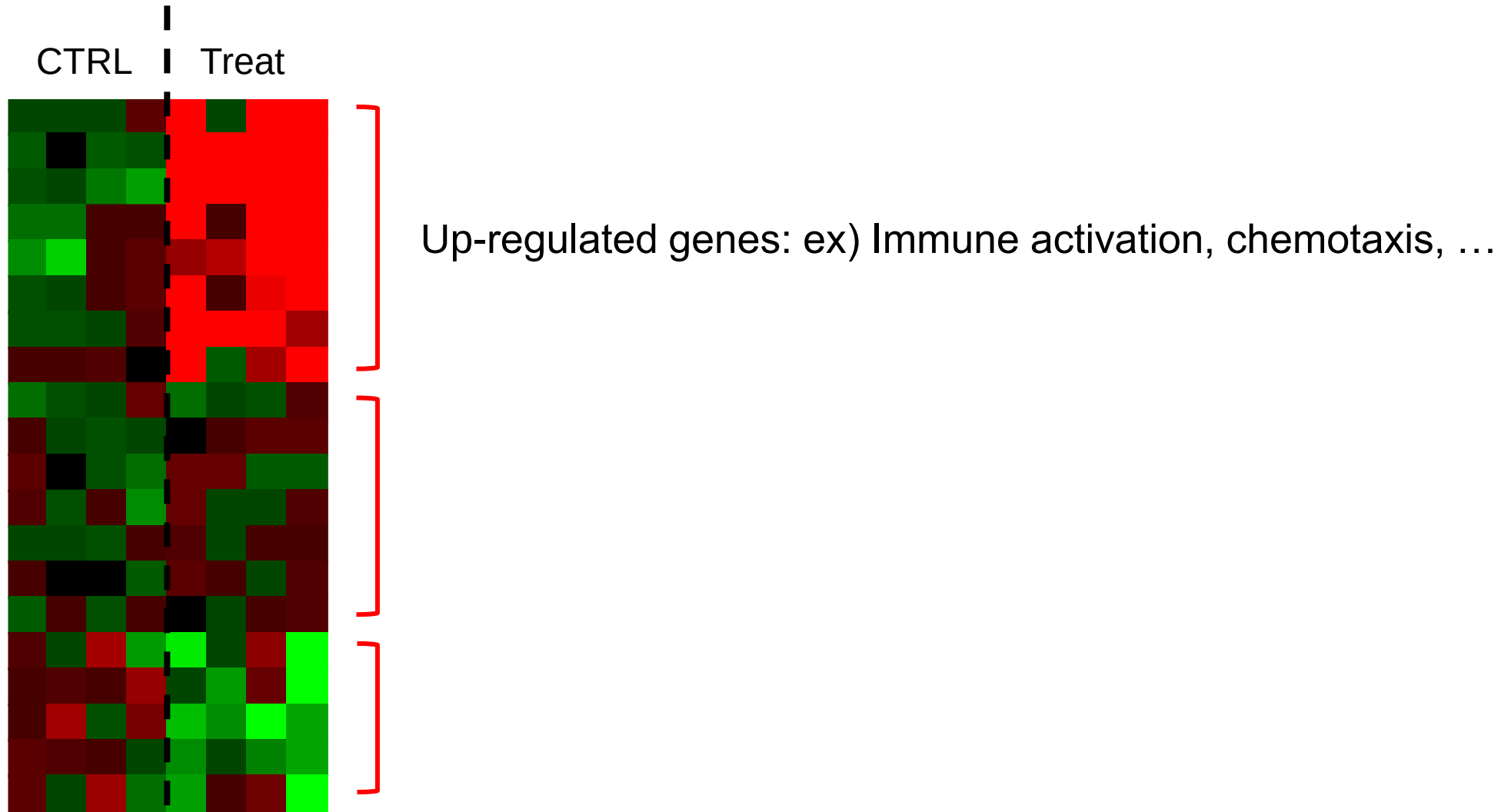
P value → Adjusted p value
- Multiple hypothesis correction
- Avoid lucky hits my multiple testing

- Gene set analysis

We can perform gene-centric analysis

But! Too many! (20k genes)

Let's see whether there is a coherent pathway between genes



- Geneset Database

Researchers already studied a lot !
We don't need to start from the scratch

Gene Ontology



MSigDB-c2



REACTOME



KEGG



BIOCARTA



HumanCyc

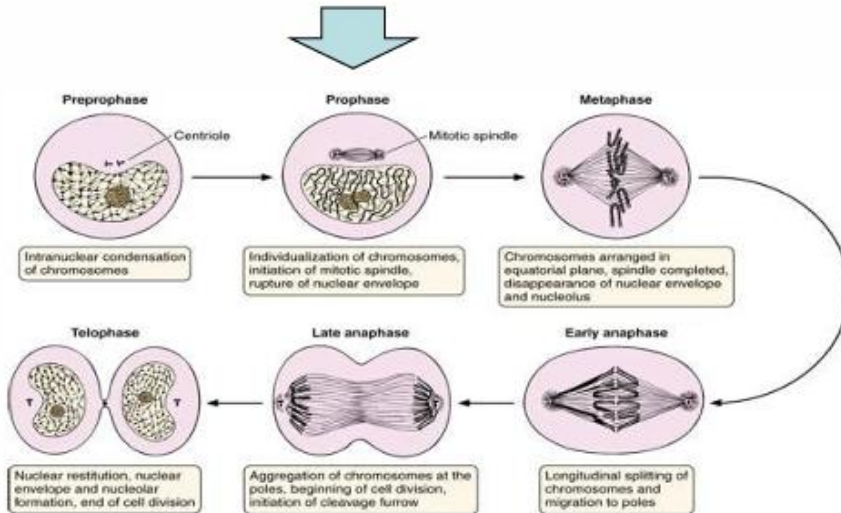
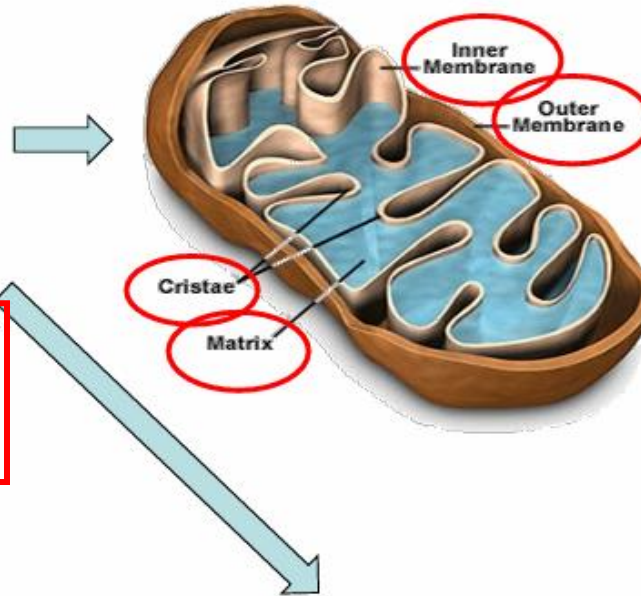


- GO (Gene ontology)

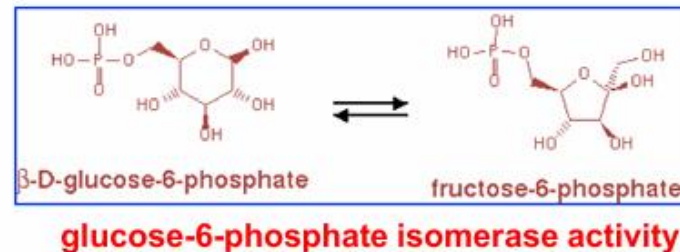
-Description about a given gene

❖ Three GO domains

- **Cellular component:** the parts of a cell or its extracellular environment;
- **Molecular function:** the elemental activities of a gene product at the molecular level, such as binding or catalysis;
- **Biological process:** operations or sets of molecular events with a defined beginning and end, pertinent to the functioning of integrated living units (cells, tissues, organs, and organisms);

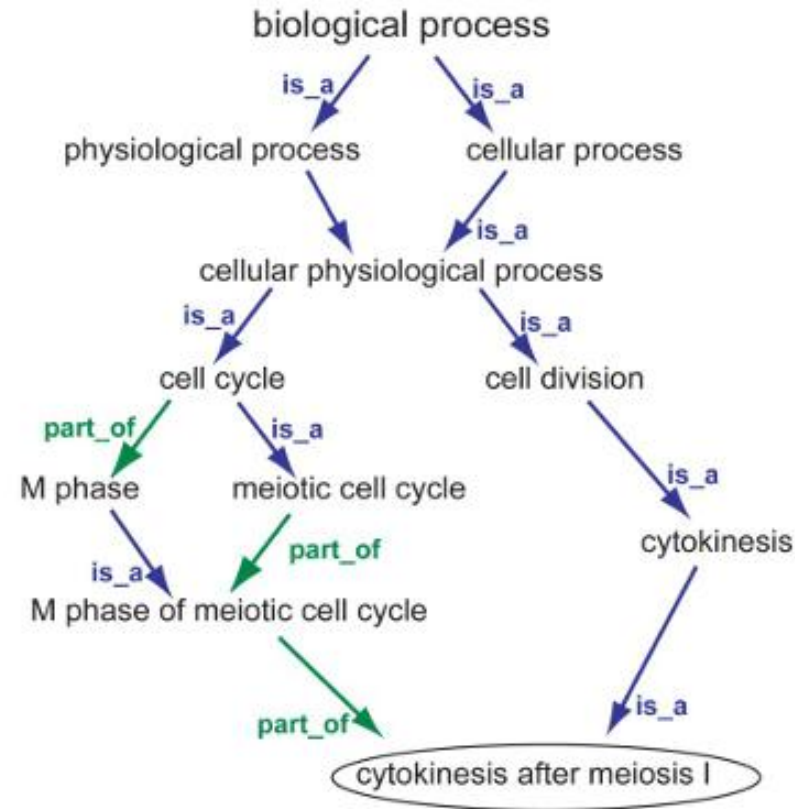


Cell division



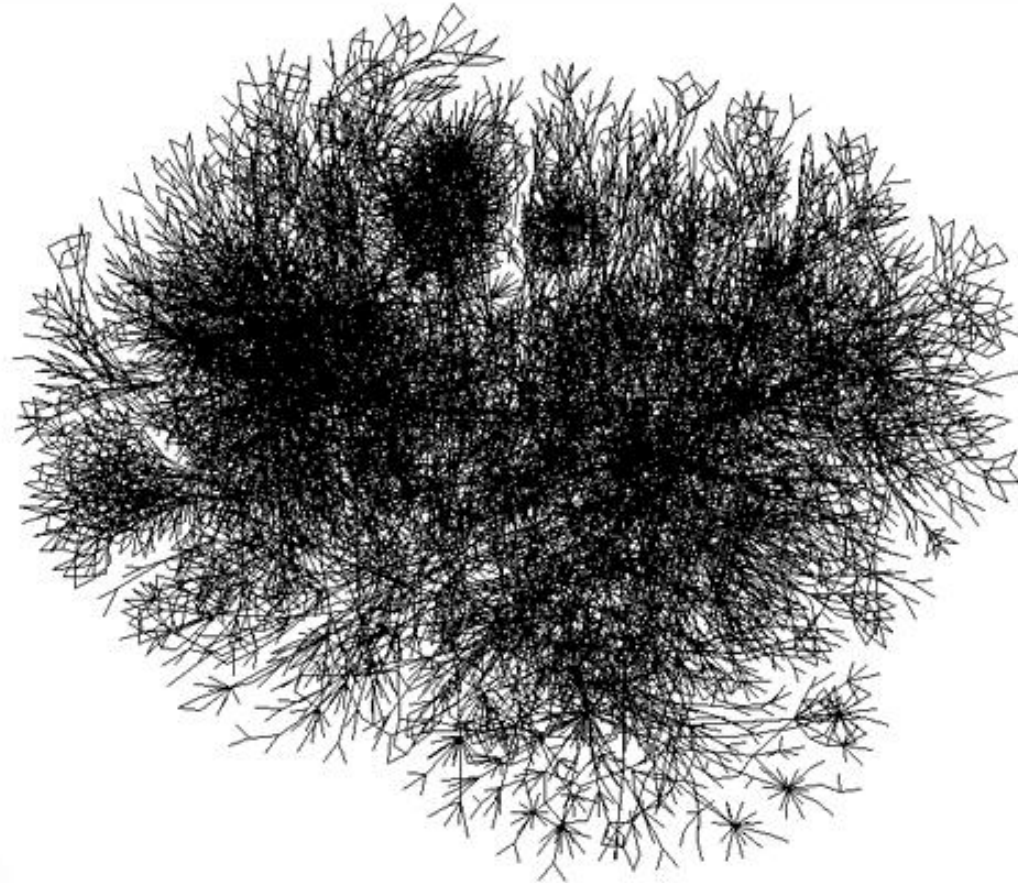
- GO (Gene ontology)

➤ Example of GO tree



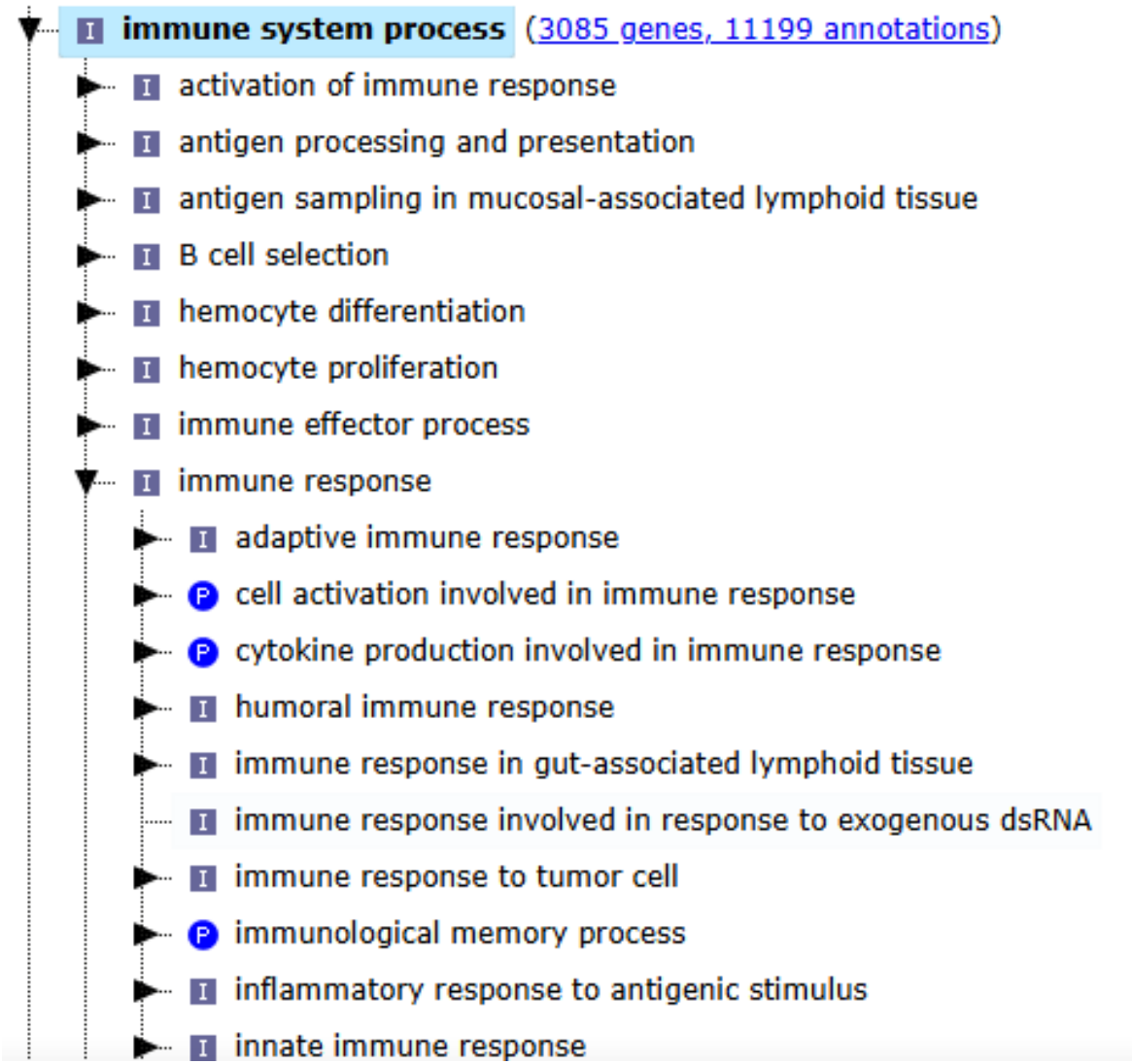
“cellular physiological process”, “M phase of meiotic cell cycle” and “cytokinesis after meiosis I” have **two parents**.

➤ Layout of whole GO structure



15,335 *is_a* or *part_of* relationships between 9,199 GO *biological process* terms (as of March 2005, by Insuk Lee)

- GO (Gene ontology)



- GO (Gene ontology); CD4-positive, alpha-beta T cell proliferation

Term ID

CD4-positive, alpha-beta T cell proliferation
GO:0035739

<< first < prev 1 next > last >>

100 ▾

Showing items 1 - 60 of 60

Export:

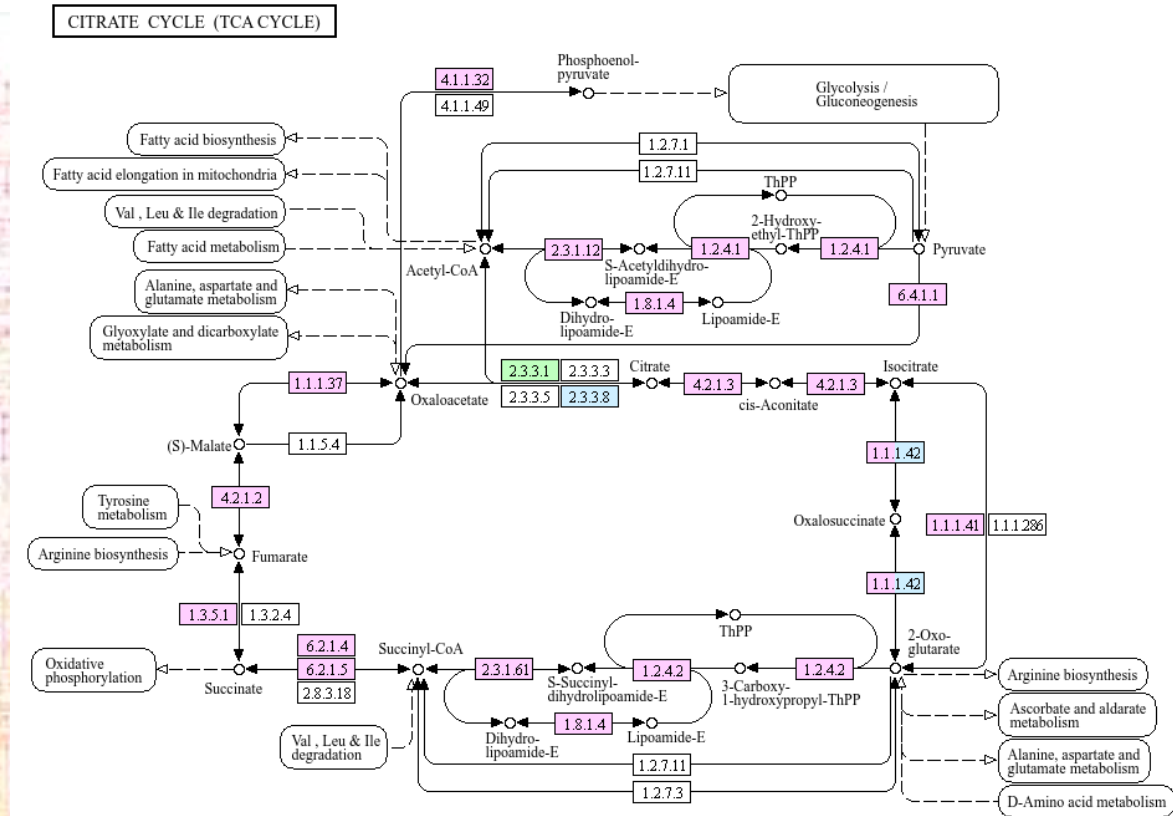
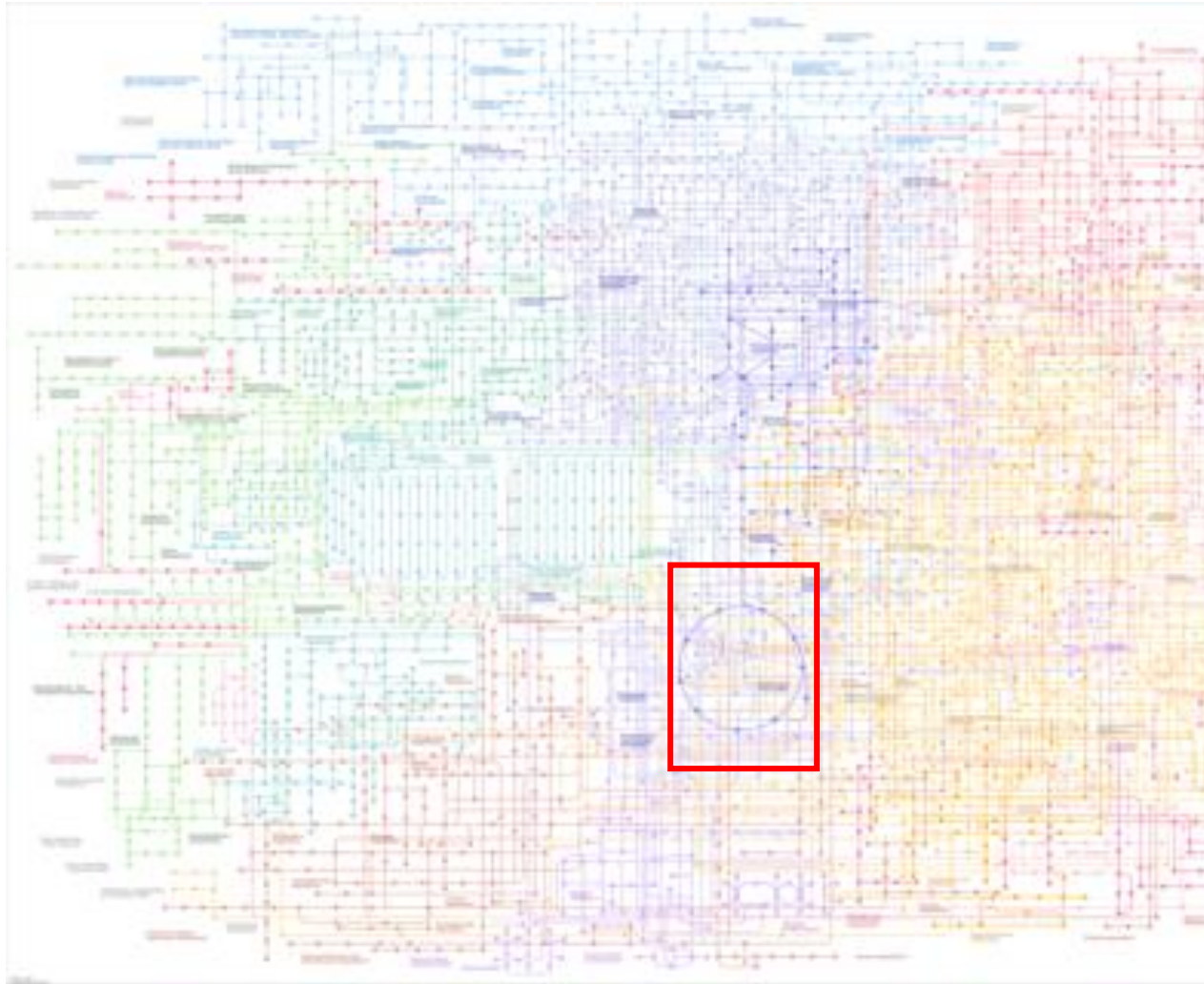
Text File

Excel File

MouseMine

Symbol, Name	Chr	Annotated Term	Context	Proteoform	Evidence	Inferred From	Reference(s)
Arg2, arginase type II	12	negative regulation of CD4-positive, alpha-beta T cell proliferation			IMP		J:243479 [PMID:25009204]
Card11, caspase recruitment domain family, member 11	5	CD4-positive, alpha-beta T cell proliferation			IMP	MGI:3039682	J:89322 [PMID:12867038]
Card11, caspase recruitment domain family, member 11	5	positive regulation of CD4-positive, alpha-beta T cell proliferation	positively regulates CD4-positive, alpha-beta T cell proliferation.		IMP	MGI:3039682	J:89322 [PMID:12867038]
Cblb, Casitas B-lineage lymphoma b	16	CD4-positive, alpha-beta T cell proliferation			IMP	MGI:2180572	J:89097 [PMID:14973438]
Cblb, Casitas B-lineage lymphoma b	16	negative regulation of CD4-positive, alpha-beta T cell proliferation	negatively regulates CD4-positive, alpha-beta T cell proliferation.		IMP	MGI:2180572	J:89097 [PMID:14973438]
Cd3e, CD3 antigen, epsilon polypeptide	9	CD4-positive, alpha-beta T cell proliferation			IDA		J:17350 [PMID:8125140]
Cd3e, CD3 antigen, epsilon polypeptide	9	CD4-positive, alpha-beta T cell proliferation			IDA		J:75401 [PMID:11894097]
Cd3e, CD3 antigen, epsilon polypeptide	9	CD4-positive, alpha-beta T cell proliferation			IDA		J:89097 [PMID:14973438]
Cd3e, CD3 antigen, epsilon polypeptide	9	positive regulation of CD4-positive, alpha-beta T cell proliferation	positively regulates CD4-positive, alpha-beta T cell proliferation.		IDA		J:17350 [PMID:8125140]
Cd3e, CD3 antigen, epsilon polypeptide	9	positive regulation of CD4-positive, alpha-beta T cell proliferation	positively regulates CD4-positive, alpha-beta T cell proliferation.		IDA		J:75401 [PMID:11894097]
		positive regulation of CD4-	positively regulates CD4-positive				

- KEGG (Kyoto Encyclopedia of Genes and Genomes)



- GMT format

regulation of cardiac conduction	GO:1903779	ATP2A1	ATP2A2	ATP2A3	ATP2B1	ATP2B2	ATP2B3	ATP2B4	PRKACA	SLC8A2	SLC8A2
epithelial cilium movement involved in extracellular fluid movement	GO:0003351	CCDC40	ADCY10								
endoplasmic reticulum tubular network membrane organization	GO:1990809	ARL6IP1	ATL1	RTN4	ATL2						
negative regulation of cilium assembly	GO:1902018	LIMK2	TESK1	CDK10	CCP110	YAP1	TBC1D30	TBC1D7	ODF2L	CEP97	TOX1
regulation of response to interferon-gamma	GO:0060330	PARP9									
histone H3-K9 demethylation	GO:0033169	KDM4A	KDM1A	KDM4B	KDM4C	PHF8	KDM4D	KDM3A	KDM7A	KDM4E	
positive regulation of epithelial cell proliferation involved in wound healing	GO:0060054	MMP12	WNT7A	FZD7	CLDN1	ODAM	LACRT				
negative regulation of protein secretion	GO:0050709	APOE	DRD2	DRD3	DRD4	IL12A	IL12B	INS	ERP29	SERGEF	RH
determination of left/right symmetry	GO:0007368	DNAH5	FOXJ1	ZIC3	DNAH11	KIF3B	DNAI1	NPHP3	ODAD2	DNAI2	O
positive regulation of granulocyte differentiation	GO:0030854	RUNX1	EVI2B	HCLS1	HAX1	LEF1	TESC				
actin filament uncapping	GO:0051695	ACTN2									
response to metal ion	GO:0010038	MT1X	MTF1	GPHN	NDRG1	NEDD4L					
cholesterol storage	GO:0010878	SOAT1									
supramolecular fiber organization	GO:0097435	BAX	BID	COL3A1	COL5A1	CST3	FKBP1A	HSP90AB1	LTBP2	MAPT	M
enzyme-directed rRNA pseudouridine synthesis	GO:0000455	DKC1	TSR3								
positive regulation of reactive oxygen species biosynthetic process	GO:1903428	ADGRB1	CYBA	RAB27A							
L-histidine import across plasma membrane	GO:1903810	SLC7A1									
synapse assembly	GO:0007416	ACHE	BDNF	CDK5	NRCAM	POU4F1	FZD5	RAB29	PCDHB5	PCDHB14	PCDHB14

• Geneset analysis

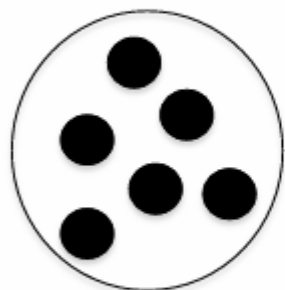
❖ *Hypergeometric test* (also known as *Fisher's exact test*)

- **Null hypothesis:** Observed list is a random sample from population.
- **Alternative hypothesis:** More black genes than expected in my list.

gene universe

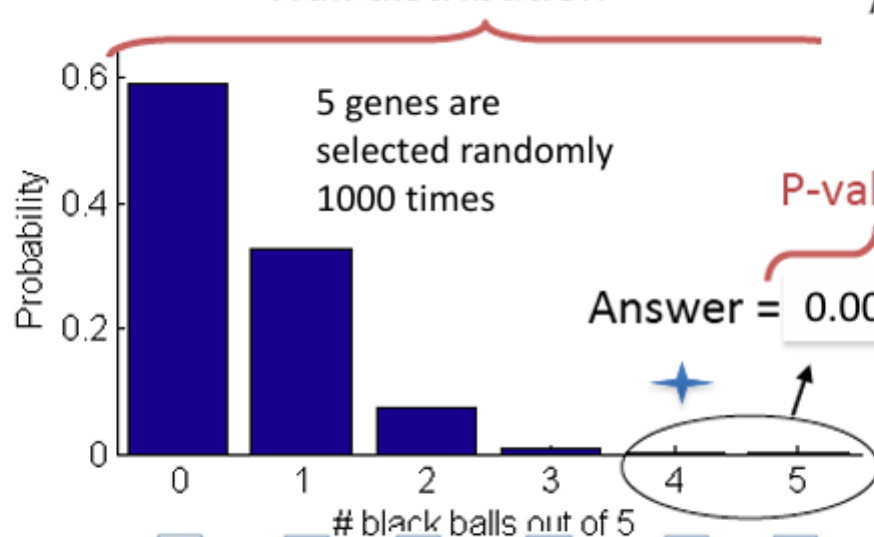


all genes in the genome or in array:
15 red, 5 black



a gene-set (e.g., for apoptosis)

Null distribution



$$P(X = x > q) = \sum_{x=q}^m \frac{\binom{m}{x} \binom{t-m}{k-x}}{\binom{t}{k}}$$

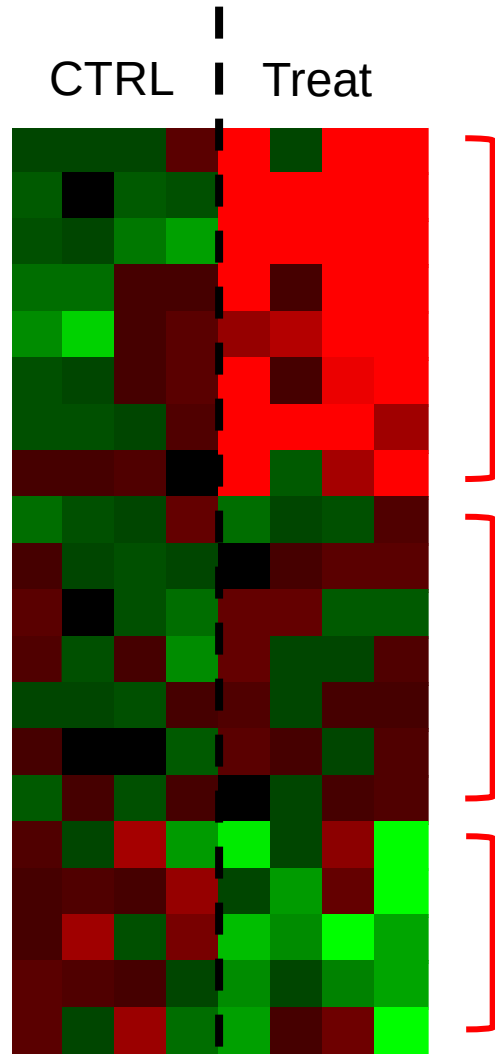
P-value

Answer = 0.001066662

If at least 4 out of 5 selected genes are black, it is very unlikely to be random ($P \sim 0.001$). There must be functional constraint for the selection of black genes.

By Quaid Morris

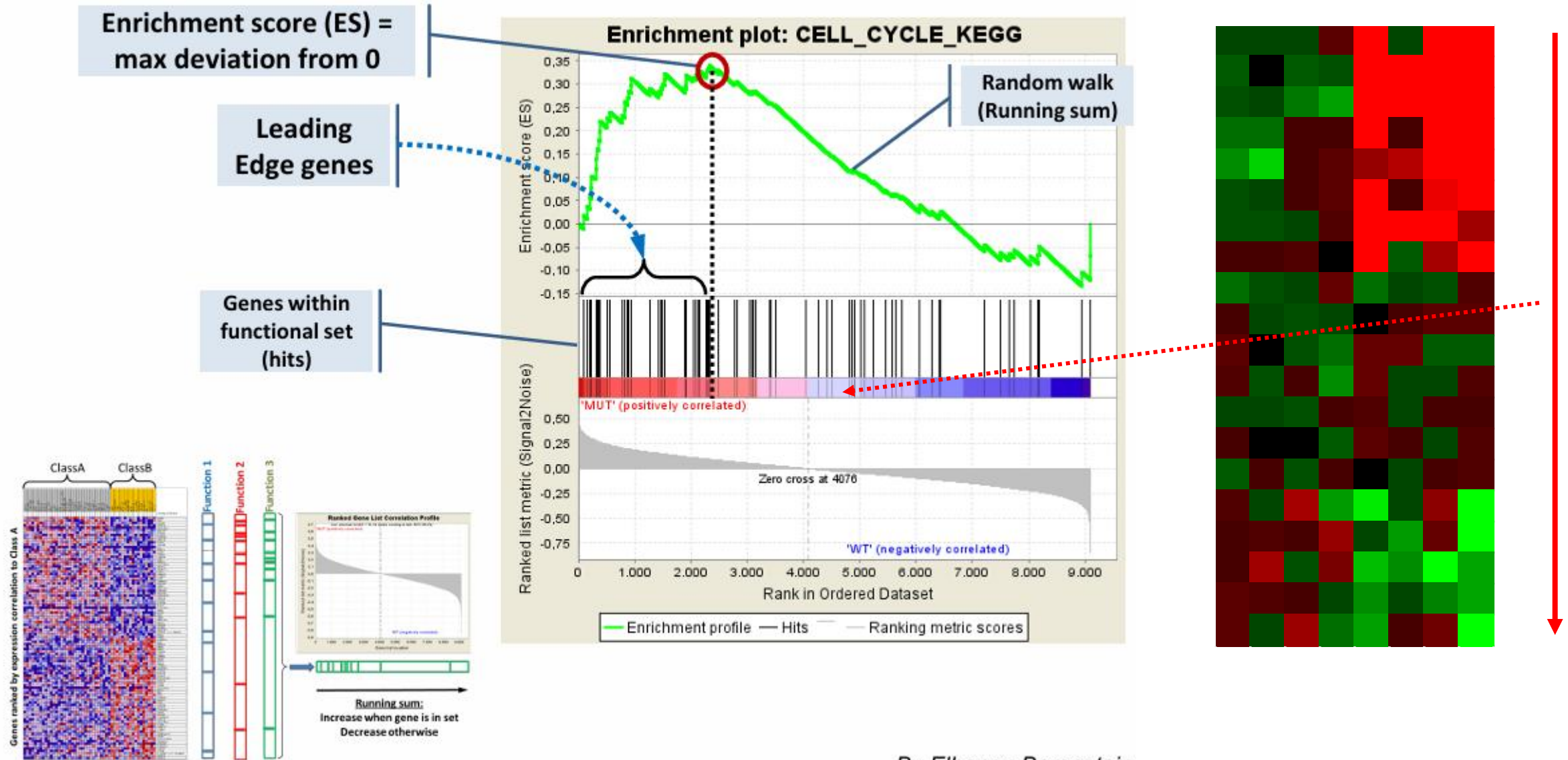
- Geneset analysis



Up-regulated genes → Fisher's exact test
→ More genes detected from "Immune activation"

- GSEA (Geneset enrichment analysis)

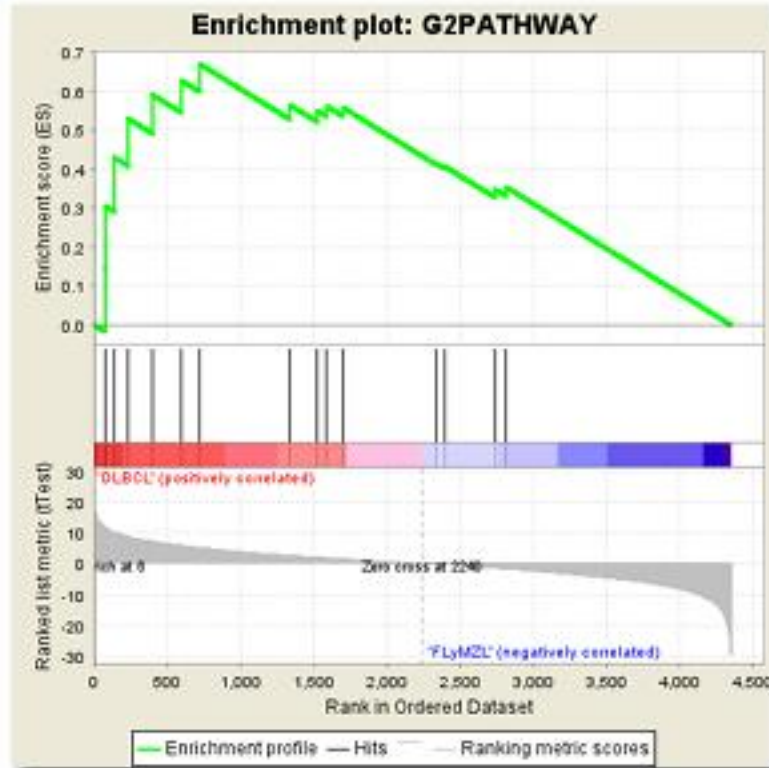
Adjusting gene weight (Fold change, Gene expression, etc)



- GSEA (Geneset enrichment analysis)

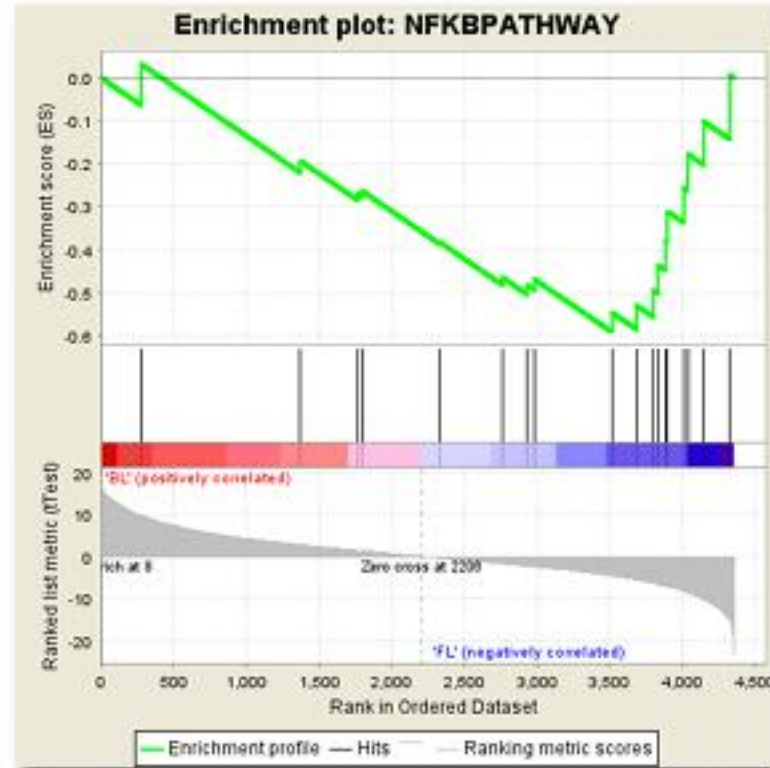
ES = 0.43

Positively enriched



ES = -0.45

Natively enriched



Low ES (no enrichment)

